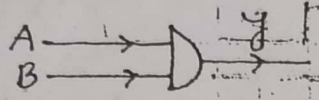
Digital Logic Design

Basic Logic Operation: There are three Basic Logic operation. They are: AND, OR, NOT

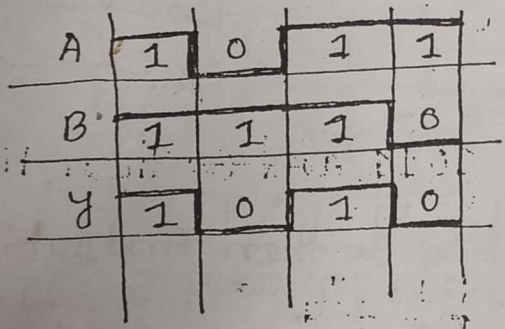
Logical Expression of AND: $y = AB$



Truth Table:

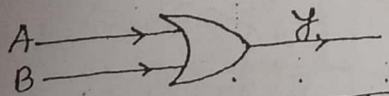
A	B	y
0	0	0
0	1	0
1	0	0
1	1	1

Timing Diagram: If A=1011 and B= 1110 then Find out the timing diagram of y.



∴ y = 1010

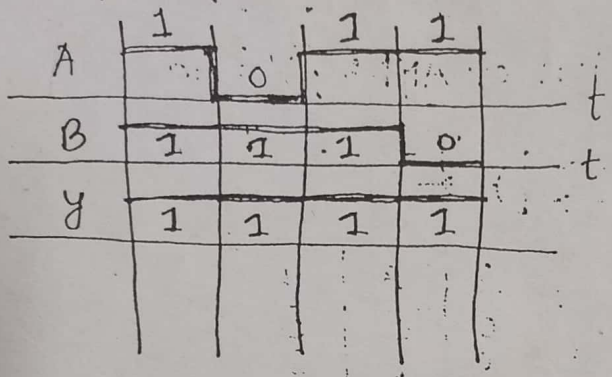
Logical Expression of OR: $y = A + B$



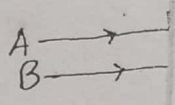
Truth Table:

A	B	y
0	0	0
0	1	1
1	0	1
1	1	1

Timing Diagram: If A=1011 and B=1110 then perform ... & to additional Logic o... Then find out the timing diagram of y.



ogical Expressio...

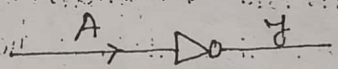


Truth Table :

Logical Expression of NOT : $y = \bar{A}$ (A')

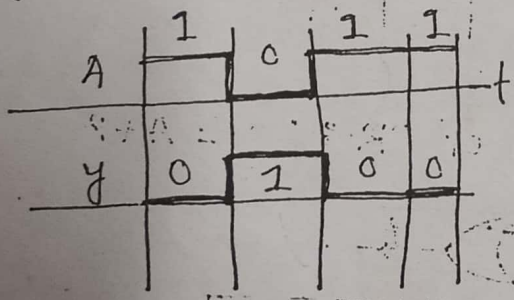
Truth Table :

A	y
0	1
1	0

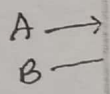


Timing Diagram...
out the timing...

Timing Diagram: If A=1011 ~~and B=~~ then find out timing diagram of y.



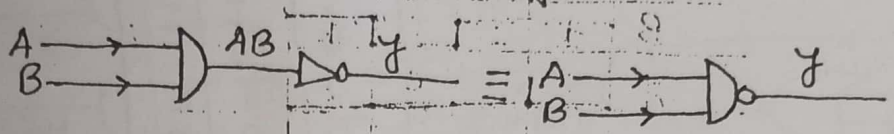
ogical Expre...



Truth Table

Additional Logic operation: There are two Additional Logic operation. They are NAND and NOR.

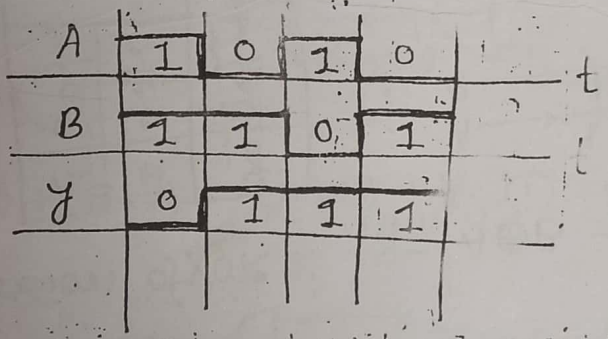
Logical Expression of NAND: $y = \overline{AB}$



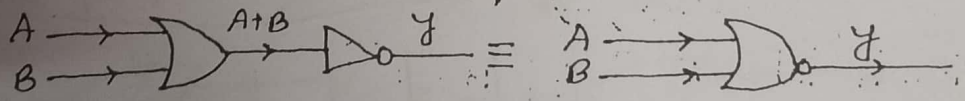
Truth Table:

A	B	y
0	0	1
0	1	1
1	0	1
1	1	0

Timing Diagram: If A = 1010 and B 1101 then find the timing diagram of y.



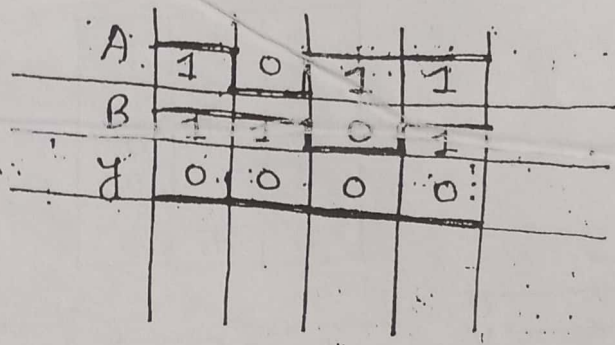
Logical Expression of NOR: $y = \overline{A+B}$



Truth Table:

A	B	y
0	0	1
0	1	0
1	0	0
1	1	0

Timing Diagram: If A = 1011 and B = 1101 then find out the timing diagram of y.

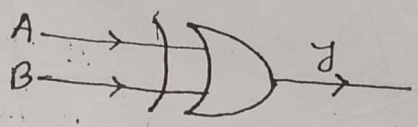


Exclusive Logic operation: There two Exclusive logic operation. They are XOR, XNOR.

Logical Expression of XOR: $y = A \oplus B = A\bar{B} + \bar{A}B$

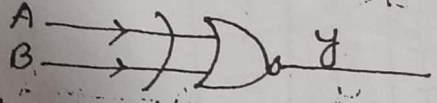
Truth Table:

A	B	y
0	0	0
0	1	1
1	0	1
1	1	0



Logical Expression of XNOR: $y = \overline{A \oplus B} = \overline{A\bar{B} + \bar{A}B}$

$= \bar{A}\bar{B} + AB$



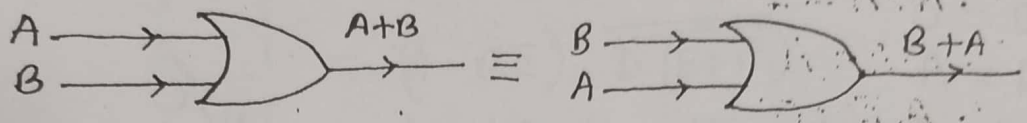
Truth Table:

A	B	y
0	0	1
0	1	0
1	0	0
1	1	1

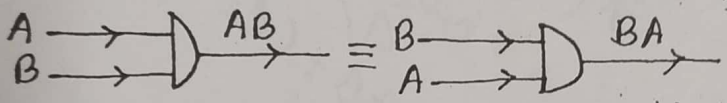
Boolean Algebra : There are 3 laws of boolean Algebra they are given below:

(i) Commutative law.
(ii) Associative law.
(iii) Distributive law.

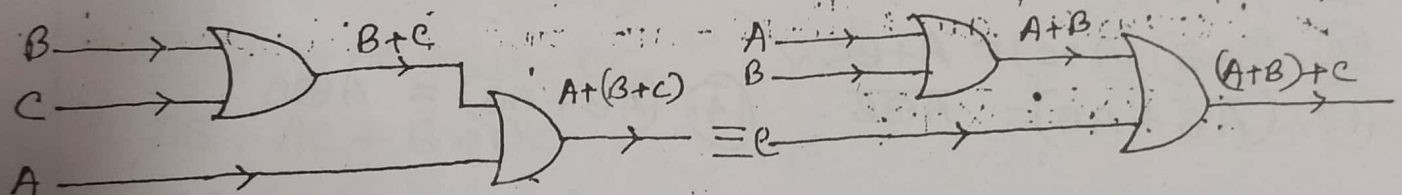
Commutative law : (i) $A+B = B+A$



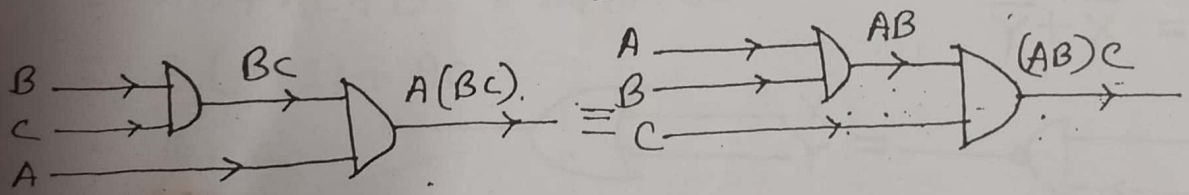
(ii) $AB = BA$



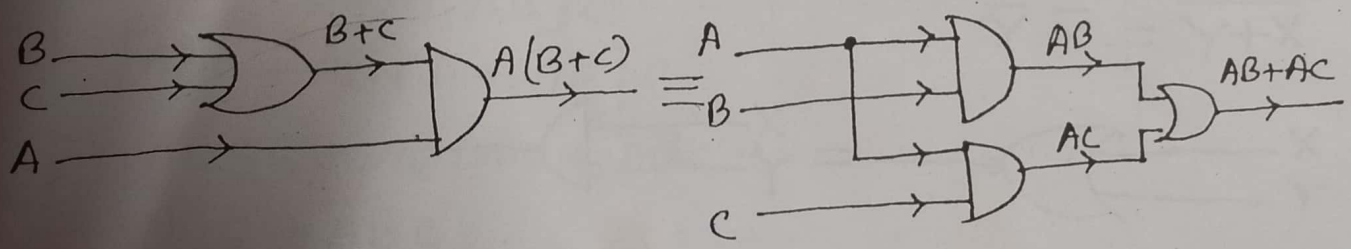
Associative law : (i) $A+(B+C) = (A+B)+C$



(ii) $A(BC) = (AB)C$



Distributive law : (i) $A(B+C) = AB+AC$

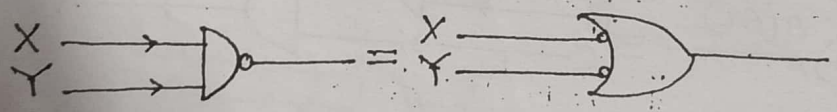


Boolean Rules: There are some Boolean rules. They are given below.

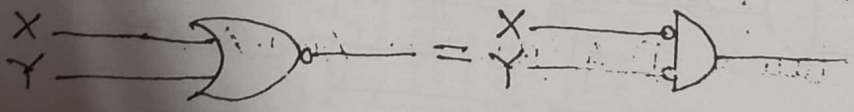
1. $A+0=A$
2. $A+1=1$
3. $A\cdot 0=0$
4. $A\cdot 1=A$
5. $A+A=A$
6. $A+\bar{A}=1$
7. $A\cdot A=A$
8. $A\cdot\bar{A}=0$
9. $A+AB=A$
10. $A+\bar{A}B=A+B$
11. $\bar{\bar{A}}=A$
12. $(A+B)(A+C)=A+BC$ ✓
13. $(\bar{A}\bar{B}+AB)=\overline{A\oplus B}$.
14. $(\bar{A}B+A\bar{B})=A\oplus B$.

De Morgan's Law: There are two De Morgan's Law.

1. $\overline{X\cdot Y}=\bar{X}+\bar{Y}$



2. $\overline{X+Y}=\bar{X}\cdot\bar{Y}$



De Morgan's Theorem :

$$\overline{(A+B)}\cdot\overline{C} + \overline{AB+\overline{D}} = \overline{(A+B)}\cdot\overline{C} \cdot \left(\overline{\overline{AB+\overline{D}}}\right)$$

$$= \left(\overline{(A+B)}+\overline{C}\right)\cdot(AB+\overline{D})$$

$$= (A+B+C)(AB+\overline{D})$$

$$\overline{A+B\overline{C}} + D\,(\overline{E+F}) = \left(\overline{A+B\overline{C}}\right)\cdot D\left(\overline{E+\overline{F}}\right)$$

$$= (A+B\overline{C})\cdot \overline{D\cdot(\overline{E+\overline{F}})}$$

$$= (A+B\overline{C})\left(\overline{D}+\left(\overline{\overline{E+\overline{F}}}\right)\right)$$

$$= (A+B\overline{C})\left(\overline{D}+E+\overline{F}\right)$$

Simplification of Boolean Expression :

(i) $AB + A(B+C) + B(B+C)$

$= AB + AB + AC + BB + BC$

$= AB + AC + B + B\overline{C}$

$= AB + AC + B(1+C)$

$= AB + AC + B \quad = AB+B+AC$

$= B(A+1) + AC$

$= B + AC$

(ii) $\left[A\overline{B}(C+BD) + \overline{A}\overline{B}\right]C$

$= \left[A\overline{B}C + A\overline{B}BD + \overline{A}\overline{B}\right]C$

$= A\overline{B}CC + A\overline{B}BDC + \overline{A}\overline{B}C$

$= A\overline{B}C + \overline{A}\overline{B}C$

$= (A+\overline{A})\,\overline{B}C$

$= \overline{B}C$

Findout the truth table $y = A + BC$.

A	B	C	BC	A+BC
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Findout the truth table $y = AB\bar{C} + A\bar{B}C + \bar{A}BC$

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

CamScanner

Standard form of Boolean expression: There are two form of Boolean expression. They are—

(i) Sum of product (SOP)

(ii) Product of sum (POS)

Convert the following SOP into SSOP: [SSOP = Standard Sum of product]

$Y = AB + ABC + \bar{B}C$

$= AB(C + \bar{C}) + ABC + (A + \bar{A})\bar{B}C$

$= ABC + AB\bar{C} + ABC + A\bar{B}C + \bar{A}\bar{B}C$

$= ABC + AB\bar{C} + A\bar{B}C + \bar{A}\bar{B}C$.

Convert the following POS into SPOS:

$Y = (A+B)(\bar{B}+C)(A+B+C)$

$= (A+B+C\bar{C})(A\bar{A}+\bar{B}+C)(A+B+C)$

$= (A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)$ [Boolean rule 11]

* SOP এর ক্ষেত্রে $A = 1$, $B = 1$; $\bar{A} = 0$, $\bar{B} = 0$

* POS এর ক্ষেত্রে $A = 0$, $B = 0$; $\bar{A} = 1$, $\bar{B} = 1$

Convert the following SOP into SPOS:

$Y = AB + \bar{B}C + \bar{A}\bar{B}\bar{C}$

$= ABC + AB\bar{C} + A\bar{B}C + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}$

$\frac{111}{7} \quad \frac{110}{6} \quad \frac{101}{5} \quad \frac{001}{1} \quad \frac{000}{0}$

$= (A+\bar{B}+C)(A+\bar{B}+\bar{C})(\bar{A}+B+C)$. [SPOS Convention]

$\frac{}{2} \quad \frac{}{3} \quad \frac{}{4}$

Convert the SSOP into SPOS :

$y = \bar{A}\bar{B}\bar{C} + AB\bar{C} + \bar{A}B\bar{C} + A\bar{B}C$ ⊗ $[0, 3, 4, 7]$

001 110 010 101

1 6 2 5

$= (A+B+C)(A+\bar{B}+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+\bar{C})$

000 011 100 111

0 3 4 7

Karnaugh Map : 3-variable karnaugh map.....

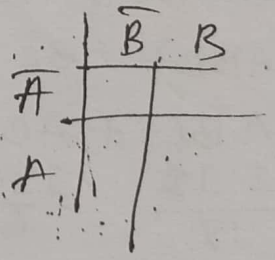
$y = A\bar{B} + B\bar{C} + AB\bar{C} + A\bar{B}\bar{C}$

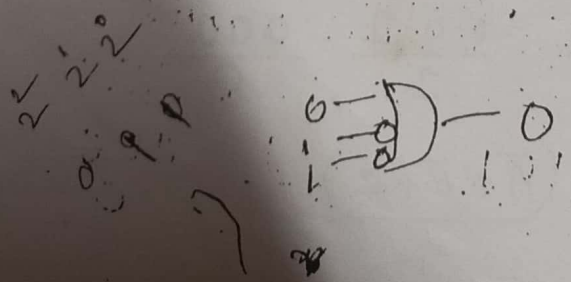
$= A\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} + \bar{A}B\bar{C} + AB\bar{C} + A\bar{B}\bar{C}$

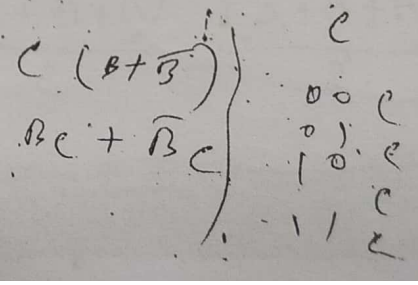
$= A\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} + \bar{A}B\bar{C}$ [∵ Any one will grante in the condition]

AB \ C	0	1	
00	0		
01	1	0	
11	1		→ Group 1.
10	1	1	→ Group 2.

∴ $y = B\bar{C} + A\bar{B}$

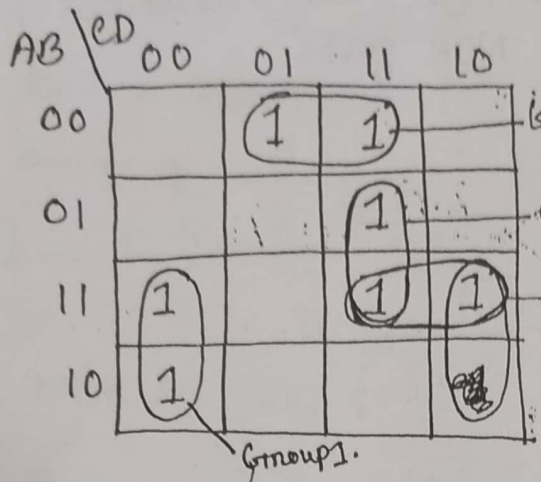






naugh Map: 4-variable karnaugh map----

$Y = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D} + ABC\bar{D} + ABCD$

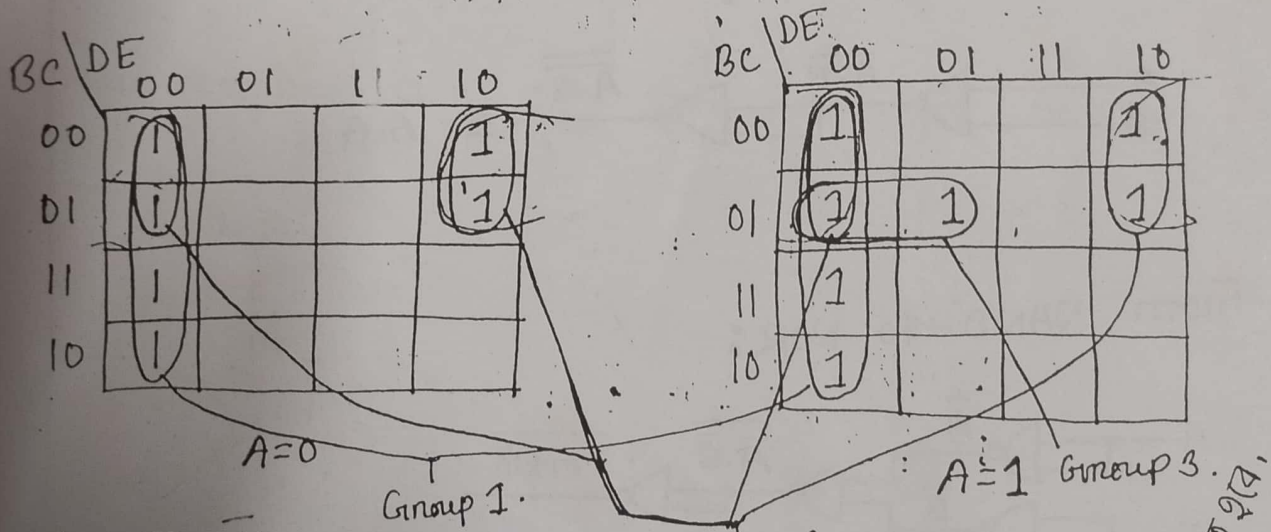


[If any one remain alone, it will join with any adjacent one]

$\therefore Y = A\bar{C}\bar{D} + \bar{A}\bar{B}D + BCD + ABC$

karnaugh map: 5-variable karnaugh map----

$Y = \bar{A}BC\bar{D}\bar{E} + A\bar{B}C\bar{D}E + \bar{A}BC\bar{D}\bar{E} + \bar{A}\bar{B}C\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}D\bar{E} + \bar{A}\bar{B}CD\bar{E} + A\bar{B}\bar{C}\bar{D}\bar{E} + A\bar{B}C\bar{D}\bar{E} + AB\bar{C}\bar{D}\bar{E} + A\bar{B}\bar{C}D\bar{E} + A\bar{B}CD\bar{E}$



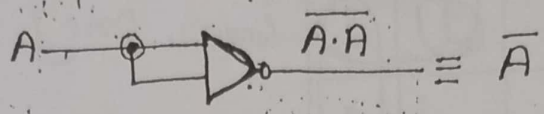
$\therefore Y = \bar{D}\bar{E} + \bar{B}\bar{D}\bar{E} + A\bar{B}C\bar{D}$

2011/13

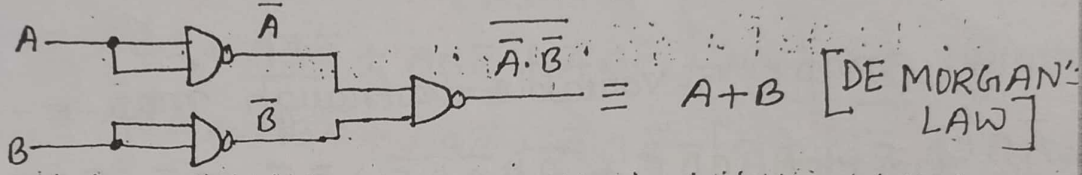
Universal gats: There are two universal gats are NAND gats and NOR gats.

(i) From NAND to NOT:

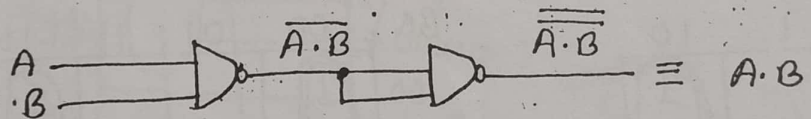
Solution



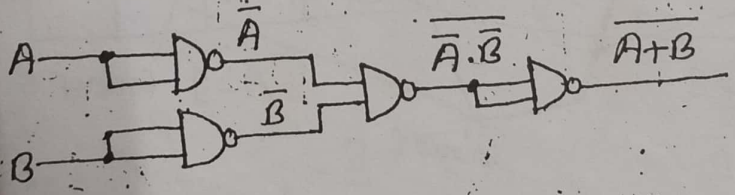
(ii) From NAND to OR:



(iii) From NAND to AND:

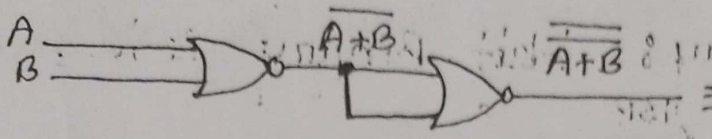


(iv) From NAND to NOR:

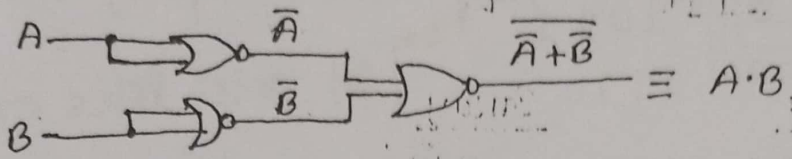


* As From NAND gats we can prove any kinds of gats, NAND gats is called Universal gats.

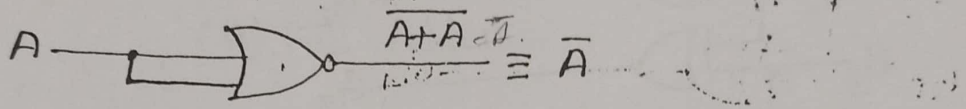
From NOR to OR:



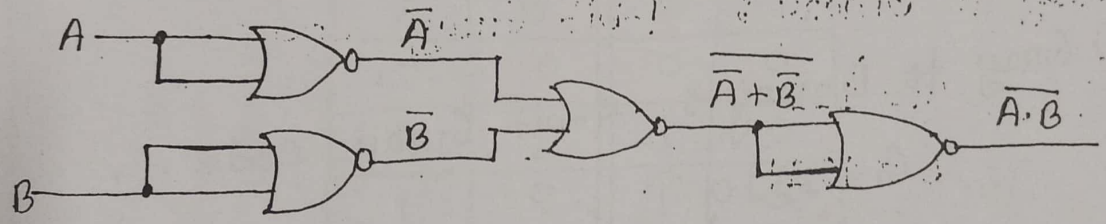
(ii) From NOR to AND:



(iii) From NOR to NOT:



(iv) From NOR to NAND:



We can express any gate from NOR gate. So NOR gate is an Universal gate.

Convertion of Binary to Gray code and vice

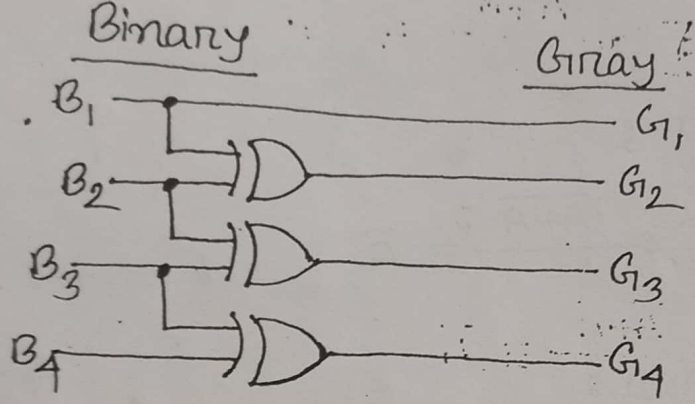
(i) Binary to Gray : 4-bit Binary code.

MSB LSB

B = 1011

G = 1110

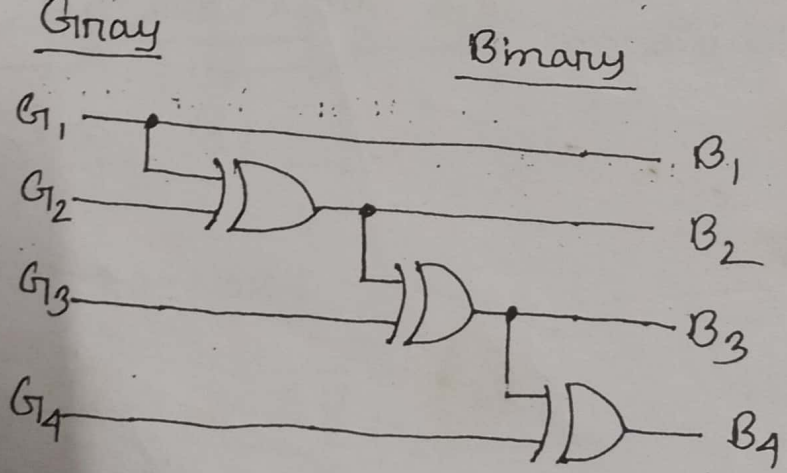
[It acts like exclusi



(ii) Gray to Binary : 4-bit Gray code.

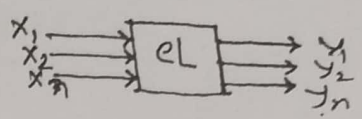
G = 1110

B = 1011

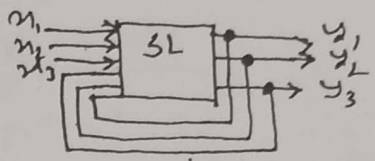


...ction of combinational logic : There are two types of functional combination logic. They are - - - - .

(i) Combinational logic. (CL)



(ii) Sequential logic (SL)



Adder : There are two adder They are —

(i) Half Adder.

(ii) Full Adder.

(i) Half adder :

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Expression :

$\therefore S = \bar{A}B + A\bar{B}$

$= A \oplus B$

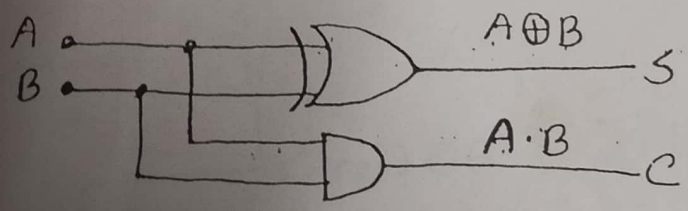
$\therefore C = A \cdot B$

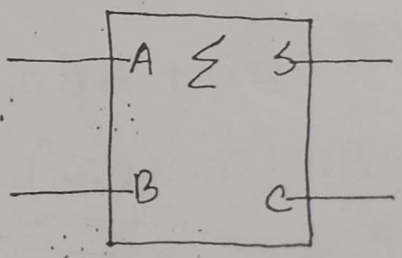
[$S = 0+0 = 0$ Cout = 0

$S = 0+1 = 1$ Cout = 0]

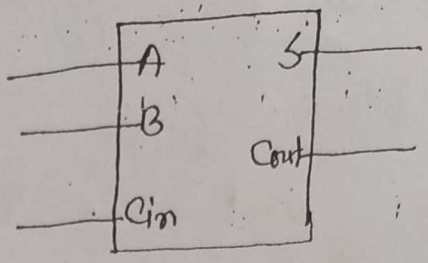
$S = 1+1 = 1$ Cout = 1]

Block diagram :





(ii) Full adder:



A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

[S বের করতে হলে At first Binary সংখ্যা গুলো Decimal যোগ করতে হয় then যোগফলকে Binary তে convert করতে হবে]

[S = 1 + 1 + 1 = 3
3 = 1 1
∴ S = 1 , C = 1]

Expression:

$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$

$= (\bar{A}B + A\bar{B})\bar{C}_{in} + (\bar{A}\bar{B} + AB)C_{in}$

$= (A \oplus B)\bar{C}_{in} + \overline{(A \oplus B)}C_{in}$ [$\because \bar{A}\bar{B} + AB = \overline{A \oplus B}$]

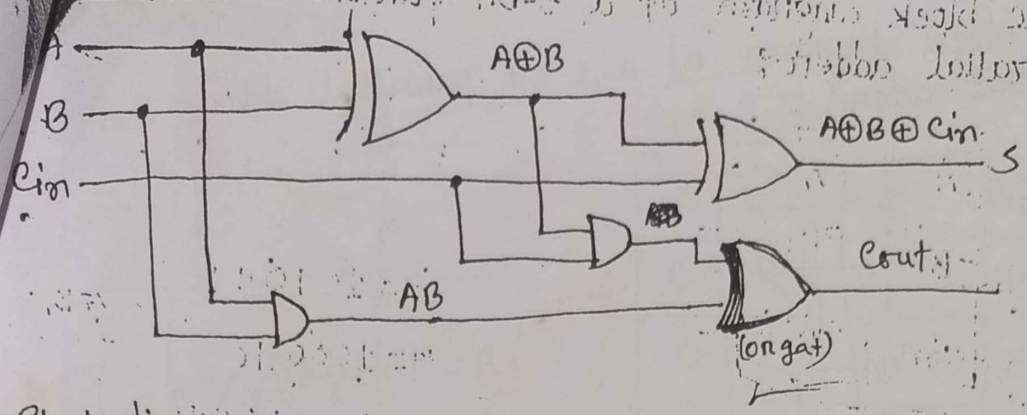
$= (A \oplus B) \oplus C_{in}$

$= A \oplus B \oplus C_{in}$

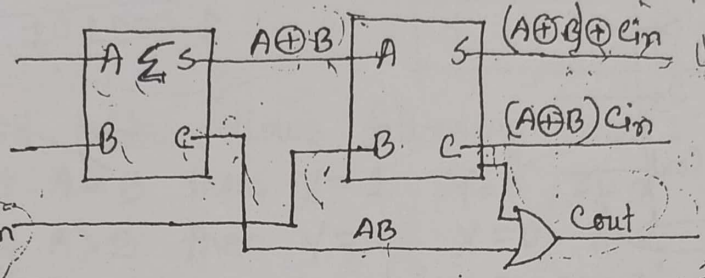
$C_{out} = AB\bar{C}_{in} + A\bar{B}C_{in} + \bar{A}BC_{in} + ABC_{in}$

$= AB(\bar{C}_{in} + C_{in}) + (\bar{A}B + A\bar{B})C_{in}$

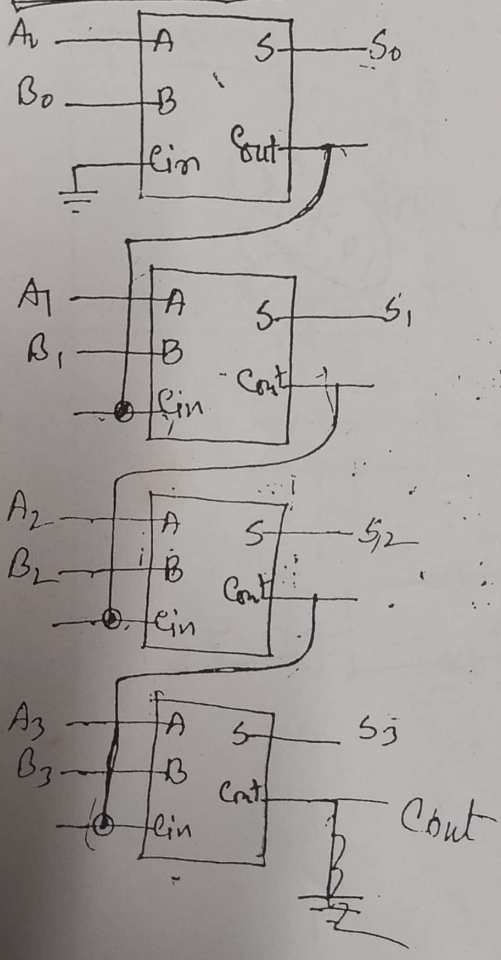
$= AB + (A \oplus B)C_{in}$



Block diagram:



Draw the 4-bit parallal adder:



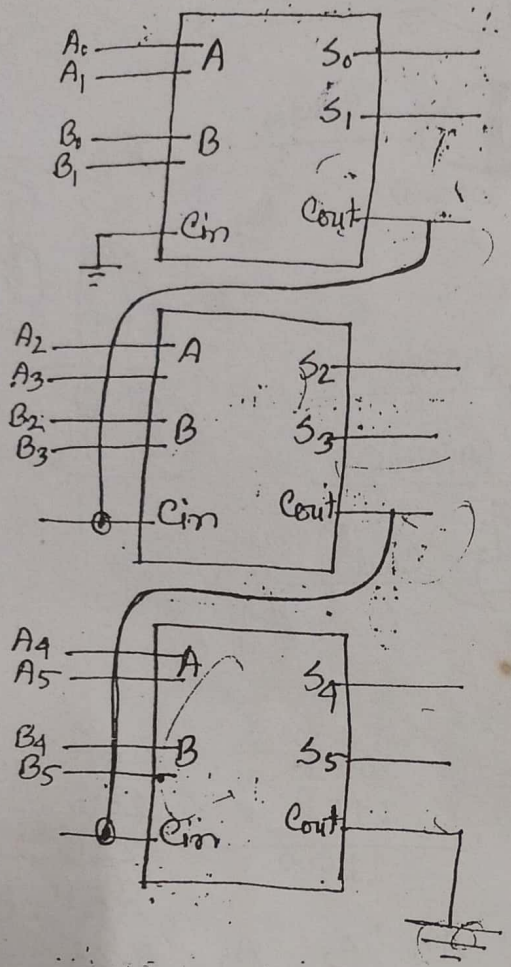
Add = 1011 (2) + 1101 (2) = 11000

② = 10
2 = 10
2 = 10
3 = 11

	A_3	A_2	A_1	A_0
A =	1	0	1	1
	B_3	B_2	B_1	B_0
B =	1	1	0	1

Cout = 1 1 0 0 0 S S_0

Draw the block diagram of a 6-bit parallal adder using 2 bit parallal adder:



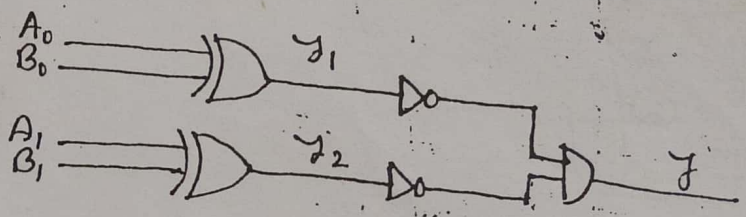
A = 101011
B = 100010
1 001101

0.25
10+15
25

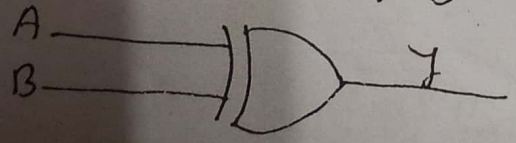
Comparator:

A	B	
0	0	= 0
0	1	= 1
1	0	= 1
1	1	= 0

$A = A_1 A_0$
$B = B_1 B_0$

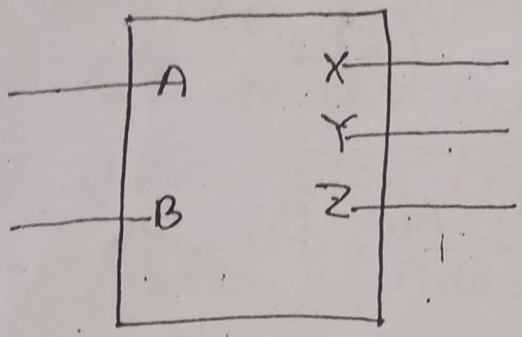


If $y = 0$ then $A = B$



else $A \neq B$.

1-bit Comparator:



Truth table:

A	B	X	Y	Z
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

This truth table follows following rules:

(i) If $A = B$ then $X = 1$, $Y = 0$, $Z = 0$

(ii) If $A > B$ then $Y = 1$, $X = 0$, $Z = 0$

(iii) If $A < B$ then $Z = 1$, $X = 0$, $Y = 0$

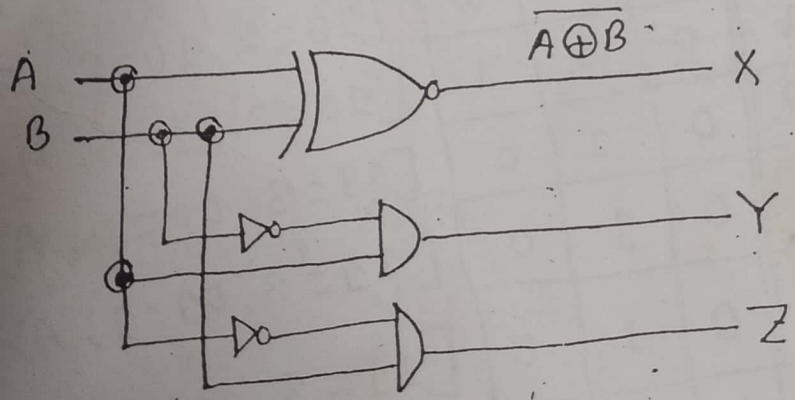
Expression:

$$X = \bar{A}\bar{B} + AB = \overline{A \oplus B}$$

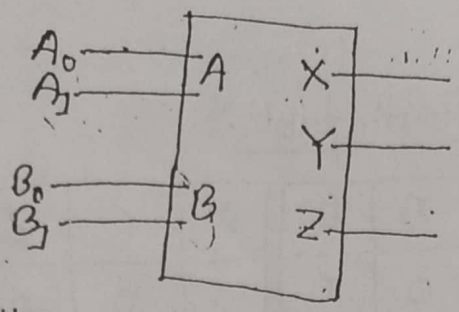
$$Y = A\bar{B}$$

$$Z = \bar{A}B$$

Circuit Diagram:



2-bit Comparator :



Truth table :

A_0	A_1	B_0	B_1	X	Y	Z
0	0	0	0	1	0	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	0	1	0
0	1	0	1	1	0	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	1	0
1	0	0	1	0	1	0
1	0	1	0	1	0	0
1	0	1	1	0	0	1
1	1	0	0	0	1	0
1	1	0	1	0	1	0
1	1	1	0	0	1	0
1	1	1	1	1	0	0

[00 = 0 00 = 0 A

[00 = 0, 01 = 1 A <

[00 = 0, 10 = 2, A <

[00 = 0, 11 = 3 A <

[01 = 1, 00 = 0 A > B]

[01 = 1, 01 = 1, A = B]

[01 = 1, 10 = 2, A < B]

[01 = 1, 11 = 3, A < B]

[10 = 2, 00 = 0, A > B]

[10 = 2, 01 = 1, A > B]

[10 = 2, 10 = 2, A = B]

[10 = 2, 11 = 3, A < B]

[11 = 3, 00 = 0, A > B]

[11 = 3, 01 = 1, A > B]

[11 = 3, 10 = 2, A > B]

[11 = 3, 11 = 3, A = B]

ression:

$$X = \bar{A}_0\bar{A}_1\bar{B}_0\bar{B}_1 + \bar{A}_0A_1\bar{B}_0B_1 + A_0\bar{A}_1B_0\bar{B}_1 + A_0A_1B_0B_1$$

$$= \bar{A}_0\bar{B}_0(\bar{A}_1\bar{B}_1 + A_1B_1) + A_0B_0(\bar{A}_1\bar{B}_1 + A_1B_1)$$

$$= (\bar{A}_1\bar{B}_1 + A_1B_1)(\bar{A}_0\bar{B}_0 + A_0B_0)$$

$$= \overline{(A_0 \oplus B_0)} \cdot \overline{(A_1 \oplus B_1)} \quad [\because \bar{A}\bar{B} + AB = \overline{A \oplus B}]$$

$$Y = \bar{A}_0A_1\bar{B}_0\bar{B}_1 + \bar{A}_0\bar{A}_1\bar{B}_0\bar{B}_1 + A_0\bar{A}_1\bar{B}_0B_1 + A_0A_1\bar{B}_0\bar{B}_1 + A_0\bar{A}_1B_0B_1 + A_0A_1B_0\bar{B}_1$$

$$= A_1\bar{B}_1(\bar{A}_0\bar{B}_0 + A_0B_0) + A_0\bar{B}_0(\bar{A}_1\bar{B}_1 + A_1B_1) + A_0\bar{A}_1\bar{B}_0B_1 + A_0A_1\bar{B}_0\bar{B}_1$$

$$= A_1\bar{B}_1\overline{(A_0 \oplus B_0)} + A_0\bar{B}_0\overline{(A_1 \oplus B_1)} + A_0\bar{B}_0(\bar{A}_1B_1 + A\bar{B}_1)$$

$$= A_1\bar{B}_1\overline{(A_0 \oplus B_0)} + A_0\bar{B}_0\overline{(A_1 \oplus B_1)} + A_0\bar{B}_0(A_1 \oplus B_1)$$

$$= A_1\bar{B}_1\overline{(A_0 \oplus B_0)} + A_0\bar{B}_0\left\{\overline{(A_1 \oplus B_1)} + (A_1 \oplus B_1)\right\}$$

$$= A_1\bar{B}_1\overline{(A_0 \oplus B_0)} + A_0\bar{B}_0.$$

$$= \bar{A}_0\bar{A}_1\bar{B}_0B_1 + \bar{A}_0\bar{A}_1B_0\bar{B}_1 + \bar{A}_0\bar{A}_1B_0B_1 + \bar{A}_0A_1B_0\bar{B}_1 + \bar{A}_0A_1B_0\bar{B}_1 + A_0\bar{A}_1B_0\bar{B}_1$$

$$= \bar{A}_1B_1(\bar{A}_0\bar{B}_0 + A_0B_0) + \bar{A}_0B_0(\bar{A}_1\bar{B}_1 + A_1B_1) + \bar{A}_0B_0(\bar{A}_1B_1 + A_1\bar{B}_1)$$

$$= \bar{A}_1B_1\overline{(A_0 \oplus B_0)} + \bar{A}_0B_0\overline{(A_1 \oplus B_1)} + \bar{A}_0B(A_1 \oplus B_1)$$

$$= \bar{A}_1B_1\overline{(A_0 \oplus B_0)} + \bar{A}_0B_0\left\{\overline{(A_1 \oplus B_1)} + (A_1 \oplus B_1)\right\}$$

$$= \bar{A}_1B_1\overline{(A_0 \oplus B_0)} + \bar{A}_0B_0$$

2-bit comparator: we can write expression with Truth table.

$A = A_0 A_1$

$B = B_0 B_1$

(i) If $A_0 = B_0$ AND $A_1 = B_1$ then $A = B$.

(ii) If $A_0 > B_0$ ~~AND $A_1 > B_1$~~ or $A_0 = B_0$ AND $A_1 > B_1$ then $A > B$

(iii) If $A_0 < B_0$ ~~AND $A_1 < B_1$~~ or $A_0 = B_0$ AND $A_1 < B_1$ then $A < B$.

A_0	A_1	B_0	B_1	X	Y	Z

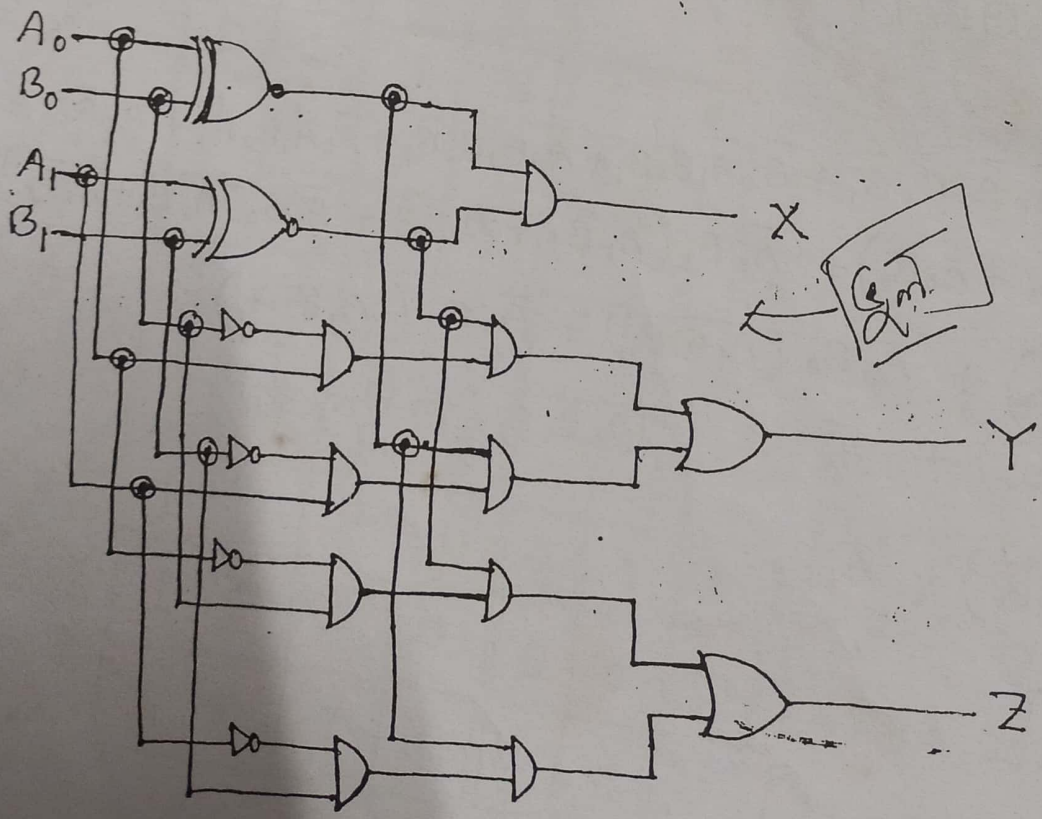
Expression:

$X = (\overline{A_0 \oplus B_0})(\overline{A_1 \oplus B_1})$

$Y = A_0 \overline{B_0}$ ~~$(\overline{A_1 \oplus B_1})$~~ $+ A_1 \overline{B_1} (\overline{A_0 \oplus B_0})$.

$Z = \overline{A_0} B_0$ ~~$(\overline{A_1 \oplus B_1})$~~ $+ \overline{A_1} B_1 (\overline{A_0 \oplus B_0})$.

Circuit Diagram:



Comparator:
A = A2 A1

Comparator:

$A = A_2 A_1 A_0$

$B = B_2 B_1 B_0$

If $A_2 = B_2$ AND $A_1 = B_1$ AND $A_0 = B_0$ then $A = B$.

If $A_2 > B_2$ OR $(A_2 = B_2$ AND $A_1 > B_1)$ OR $(A_2 = B_2$ AND $A_1 = B_1$ AND $A_0 > B_0)$ then $A > B$.

If $A_2 < B_2$ OR $(A_2 = B_2$ AND $A_1 < B_1)$ OR $(A_2 = B_2$ AND $A_1 = B_1$ AND $A_0 < B_0)$ then $A < B$.

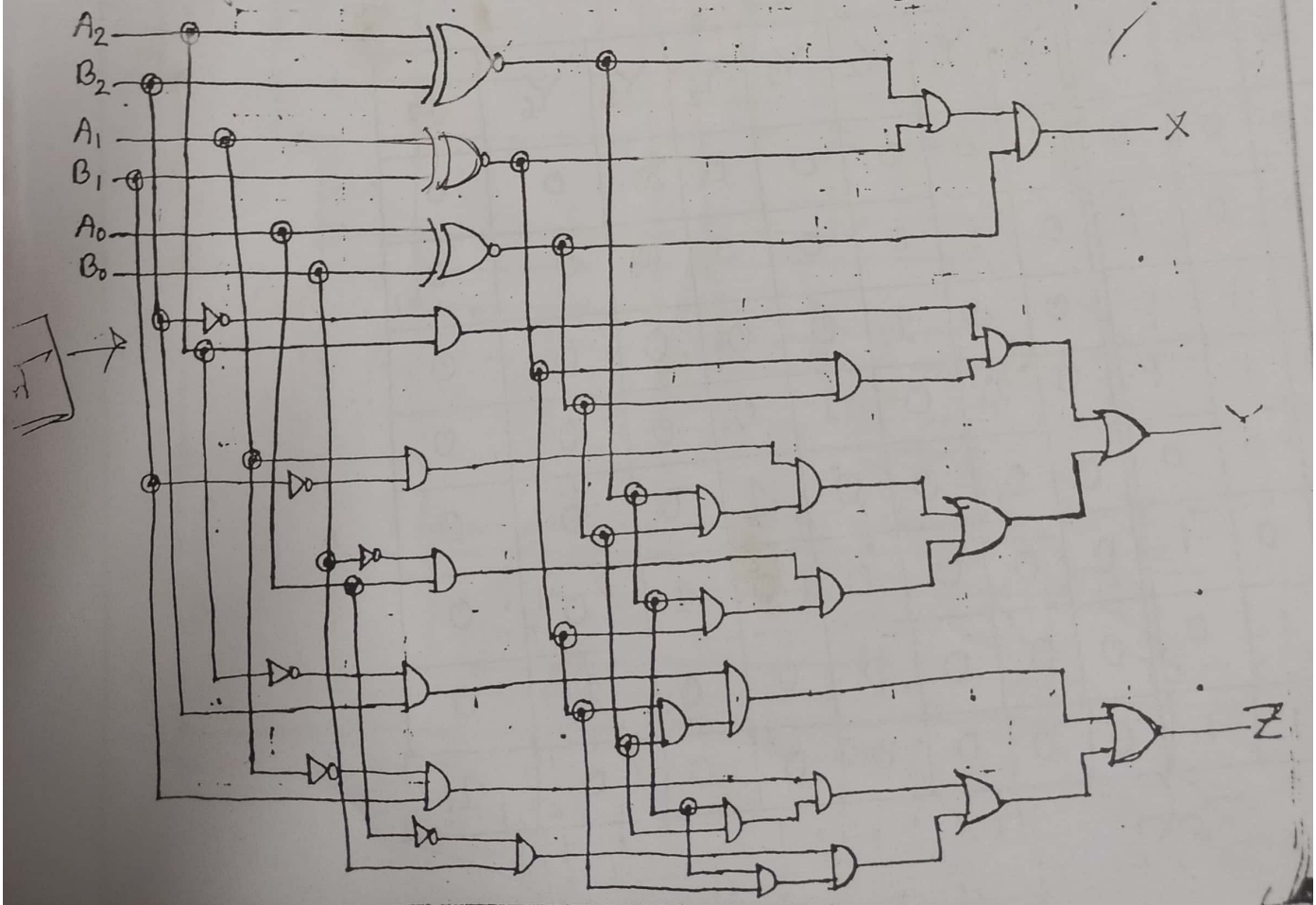
Expression:

$$X = (\overline{A_2 \oplus B_2}) \cdot (\overline{A_1 \oplus B_1}) \cdot (\overline{A_0 \oplus B_0})$$

$$Y = A_2 \overline{B_2} + A_1 \overline{B_1} (\overline{A_2 \oplus B_2}) + A_0 \overline{B_0} (\overline{A_2 \oplus B_2}) (\overline{A_1 \oplus \quad})$$

$$Z = \overline{A_2} B_2 + \overline{A_1} B_1 (\overline{A_2 \oplus B_2}) \cdot (\overline{A_2 \oplus B_2}) + \overline{A_0} B_0 (\overline{A_2 \oplus B_2}) (\overline{A_1 \oplus B_1})$$

Circuit Diagram:

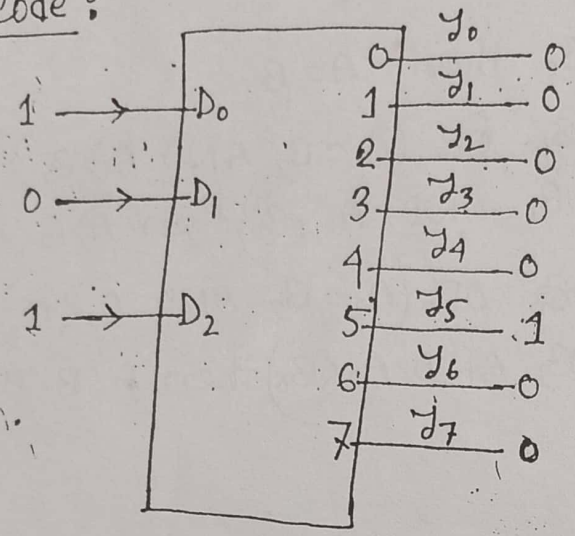


Decoder :

Encoder :

5 ⟷ 101

Code :



3-bit decoder :

Truth table :

D_0	D_1	D_2	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Expression:

$Y_0 = \bar{D}_0 \bar{D}_1 \bar{D}_2$

$Y_1 = \bar{D}_0 \bar{D}_1 D_2$

$Y_2 = \bar{D}_0 D_1 \bar{D}_2$

$Y_3 = \bar{D}_0 D_1 D_2$

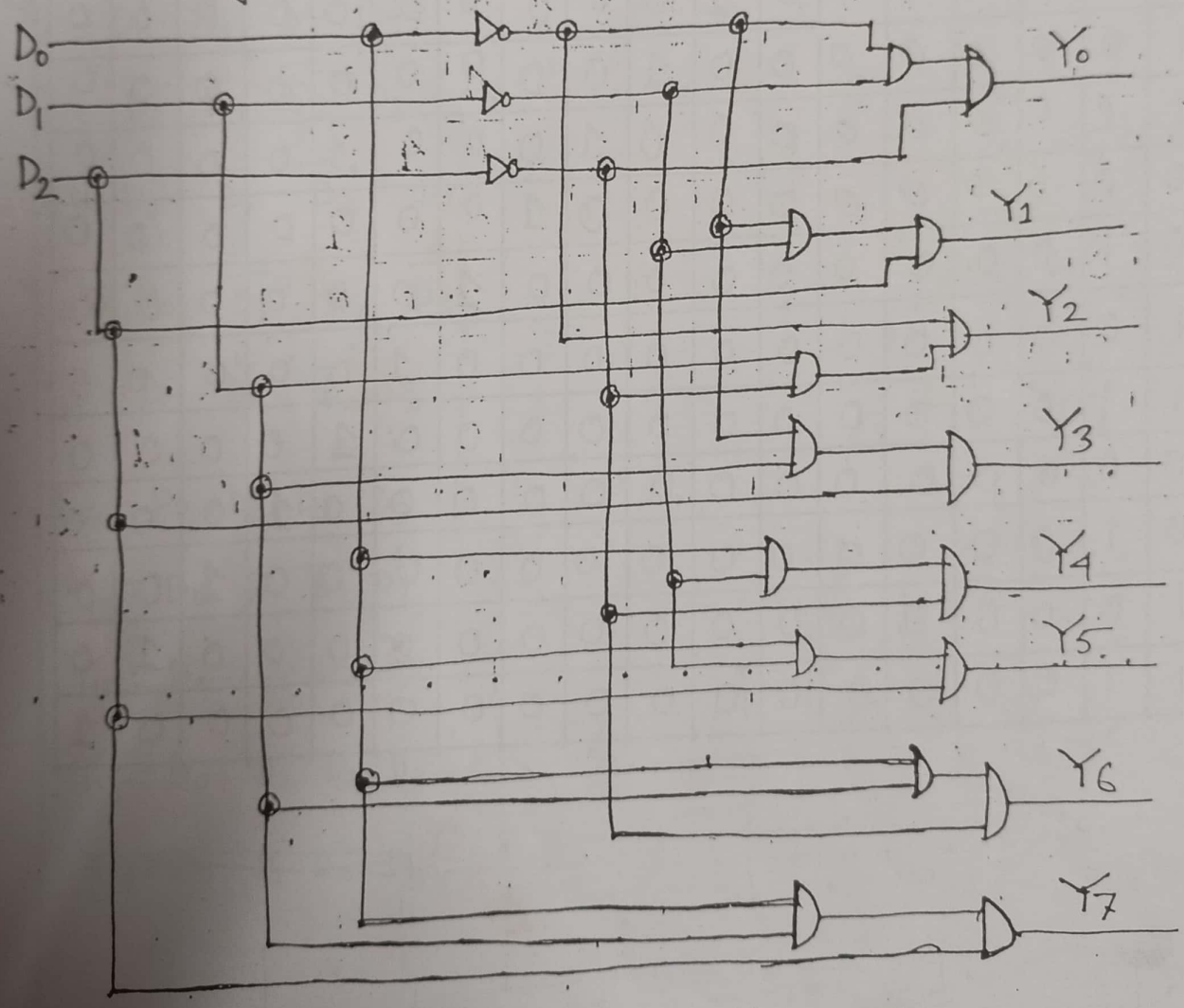
$Y_4 = D_0 \bar{D}_1 \bar{D}_2$

$Y_5 = D_0 \bar{D}_1 D_2$

$Y_6 = D_0 D_1 \bar{D}_2$

$Y_7 = D_0 D_1 D_2$

Circuit Diagram:



4-bit decoder:

Truth table:

D_0	D_1	D_2	D_3	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	Y_8	Y_9	Y_{10}	Y_{11}	Y_{12}	Y_{13}	Y_{14}	Y_{15}
0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

pression:

$Y_0 = \bar{D}_0\bar{D}_1\bar{D}_2\bar{D}_3$

$Y_1 =$

CamScanner

Expression:

$Y_0 = \overline{D_0}\,\overline{D_1}\,\overline{D_2}\,\overline{D_3}$

$Y_1 = \overline{D_0}\,\overline{D_1}\,\overline{D_2}\,D_3$

$Y_2 = \overline{D_0}\,\overline{D_1}\,D_2\,\overline{D_3}$

$Y_3 = \overline{D_0}\,\overline{D_1}\,D_2\,D_3$

$Y_4 = \overline{D_0}\,D_1\,\overline{D_2}\,\overline{D_3}$

$Y_5 = \overline{D_0}\,D_1\,\overline{D_2}\,D_3$

$Y_6 = \overline{D_0}\,D_1\,D_2\,\overline{D_3}$

$Y_7 = \overline{D_0}\,D_1\,D_2\,D_3$

$Y_8 = D_0\,\overline{D_1}\,\overline{D_2}\,\overline{D_3}$

$Y_9 = D_0\,\overline{D_1}\,\overline{D_2}\,D_3$

$Y_{10} = D_0\,\overline{D_1}\,D_2\,\overline{D_3}$

$Y_{11} = D_0\,\overline{D_1}\,D_2\,D_3$

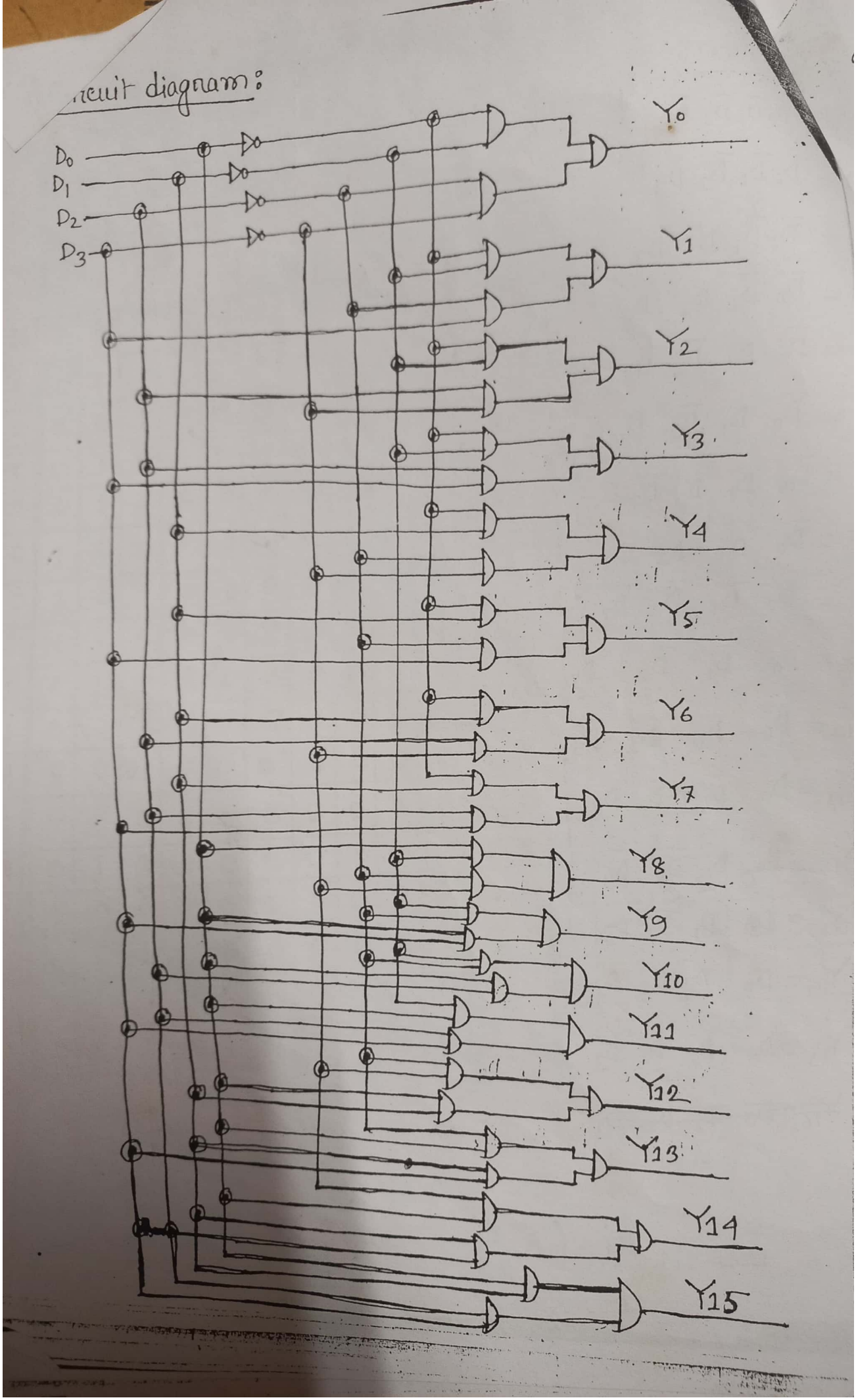
$Y_{12} = D_0\,D_1\,\overline{D_2}\,\overline{D_3}$

$Y_{13} = D_0\,D_1\,\overline{D_2}\,D_3$

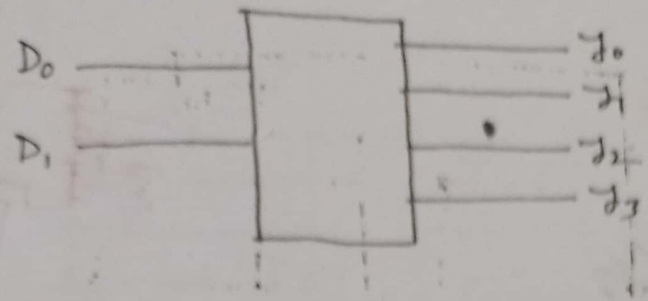
$Y_{14} = D_0\,D_1\,D_2\,\overline{D_3}$

$Y_{15} = D_0\,D_1\,D_2\,D_3$

~~$Y_{16} = D_0\,D_1\,D_2\,D_3$~~


Circuit diagram:
D0
D1
D2
D3
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
Y8
Y9
Y10
Y11
Y12
Y13
Y14
Y15

2-Bit Decoder:



Truth table:

D_0	D_1	y_0	y_1	y_2	y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Expression:

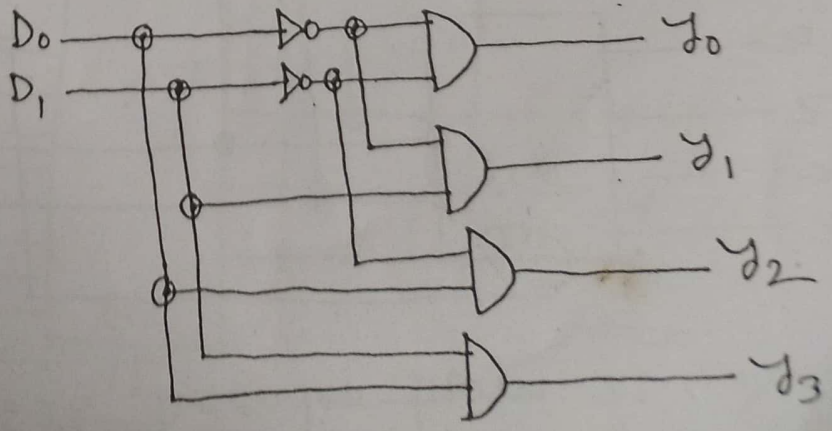
$$y_0 = \overline{D_0}\,\overline{D_1}$$

$$y_1 = \overline{D_0}\,D_1$$

$$y_2 = D_0\,\overline{D_1}$$

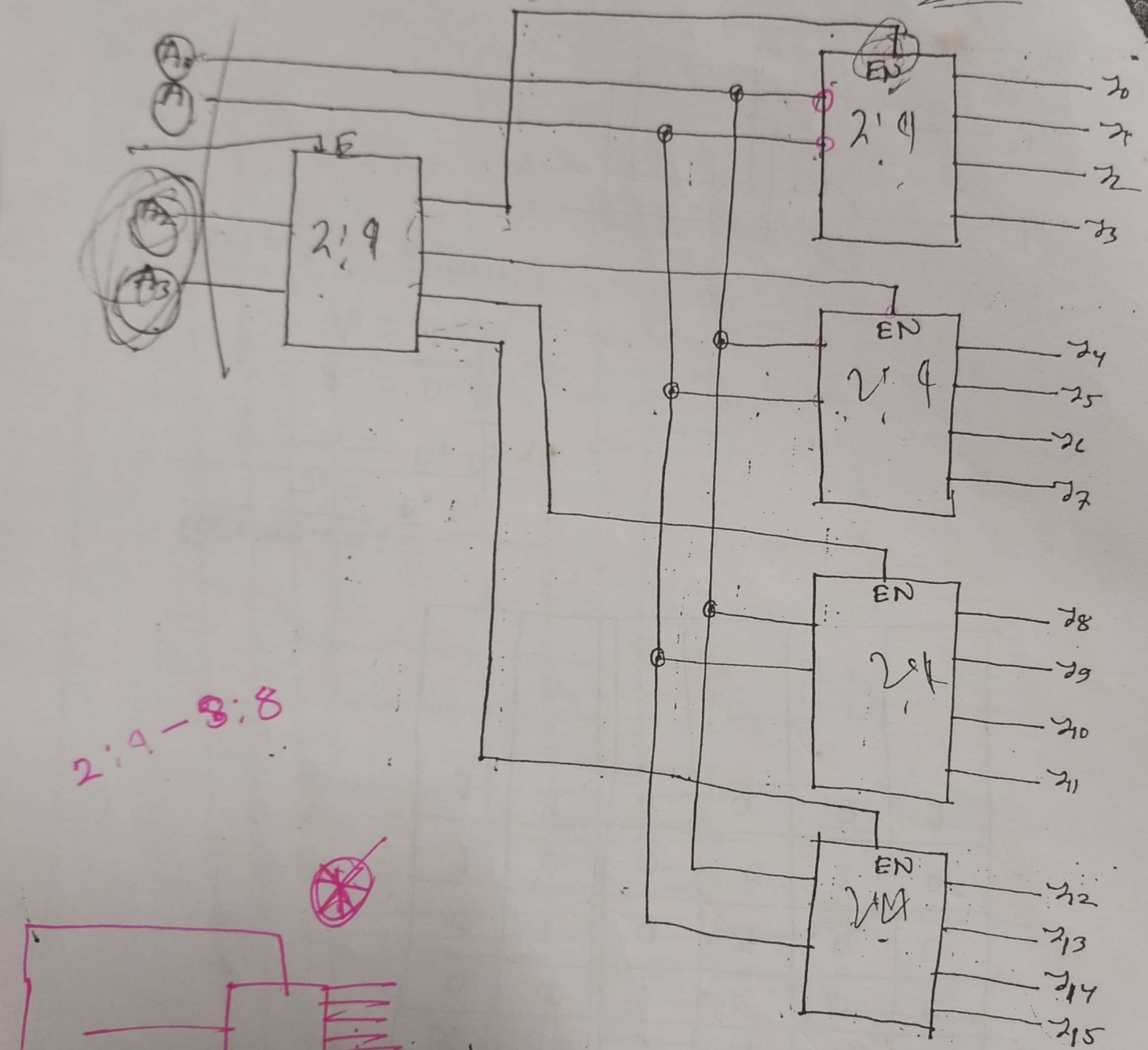
$$y_3 = D_0\,D_1$$

Circuit Diagram:

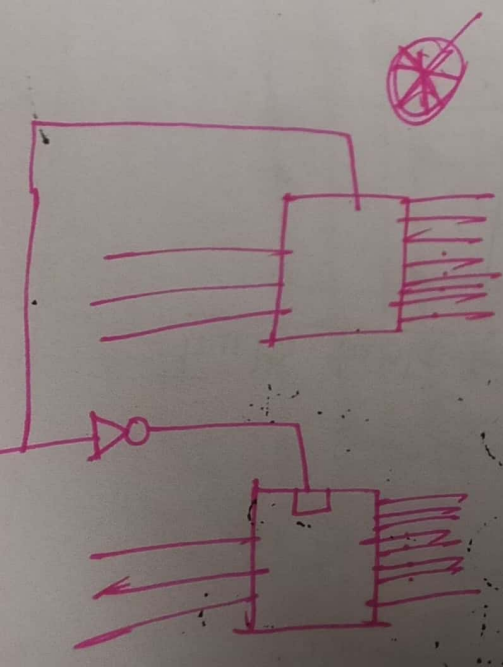


CamScanner

Determine 4:16 Decoder from 2:4 Dec

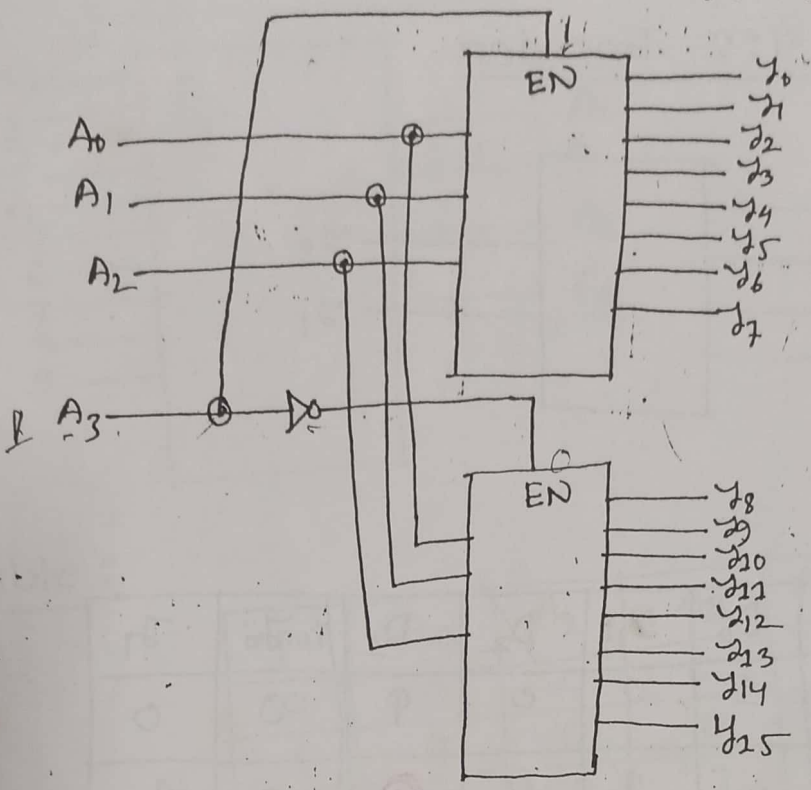


2:4 - 3:8

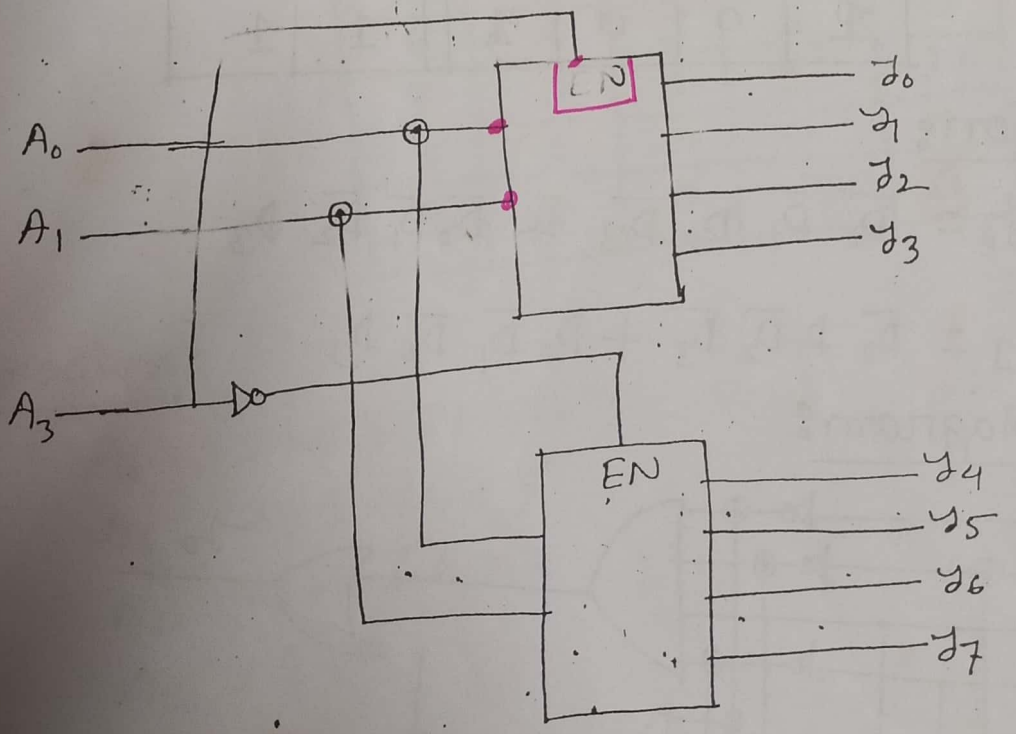


Sapti

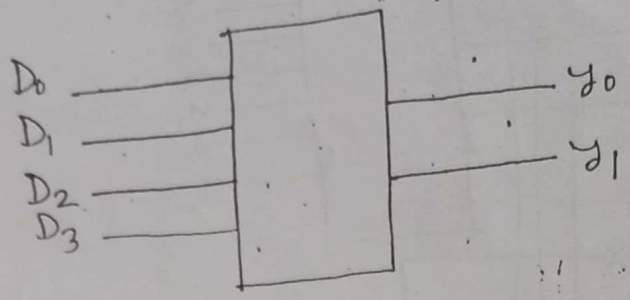
Determine 4:16 Decoder from 3:8 Decoder.



Determine 3:8 Decoder from 2:4 Decoder.



Encoder: 4:2 Encoder



Truth table:

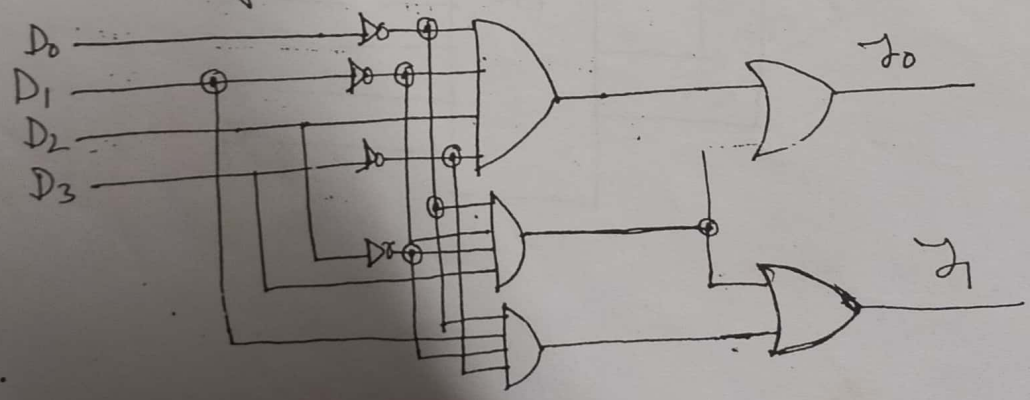
D_0	D_1	D_2	D_3	y_0	y_1
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

Expression:

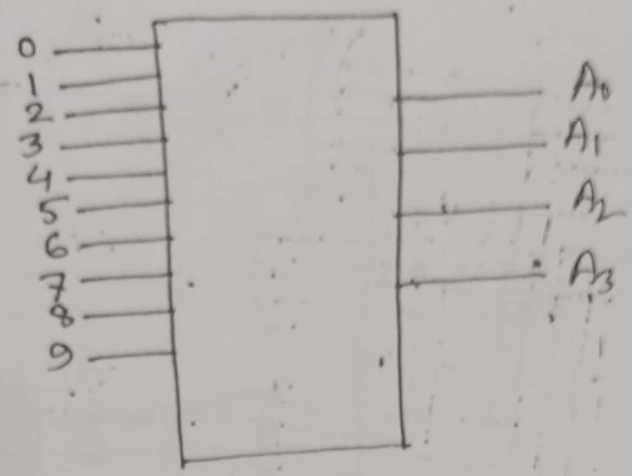
$$y_0 = \overline{D_0}\,\overline{D_1}\,D_2\,\overline{D_3} + \overline{D_0}\,\overline{D_1}\,\overline{D_2}\,D_3$$

$$y_1 = \overline{D_0}\,D_1\,\overline{D_2}\,\overline{D_3} + \overline{D_0}\,\overline{D_1}\,\overline{D_2}\,D_3$$

Circuit Diagram:



10:4 ~~Encoder~~



Truth table:

input	A_3	A_2	A_1	A_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

Expression:

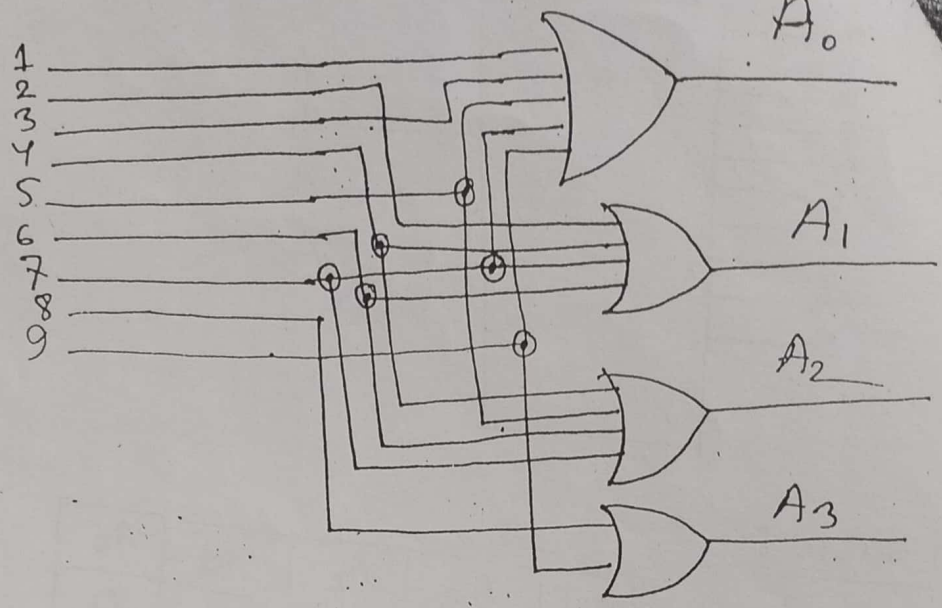
$A_0 = 1+3+5+7+9$

$A_1 = 2+3+6+7$

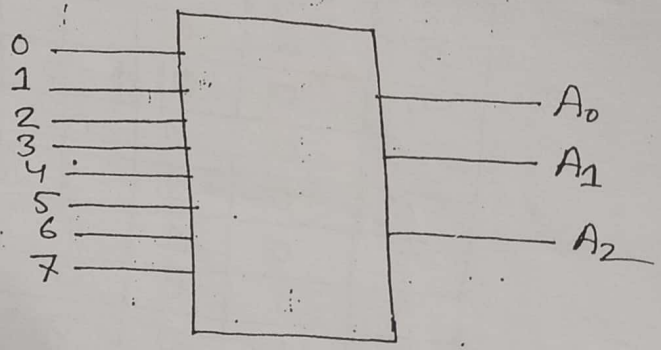
$A_2 = 4+5+6+7$

$A_3 = 8+9$

Circuit Diagram:



8:3 Encoder



Truth table:

input	A_2	A_1	A_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8			

CamScanner

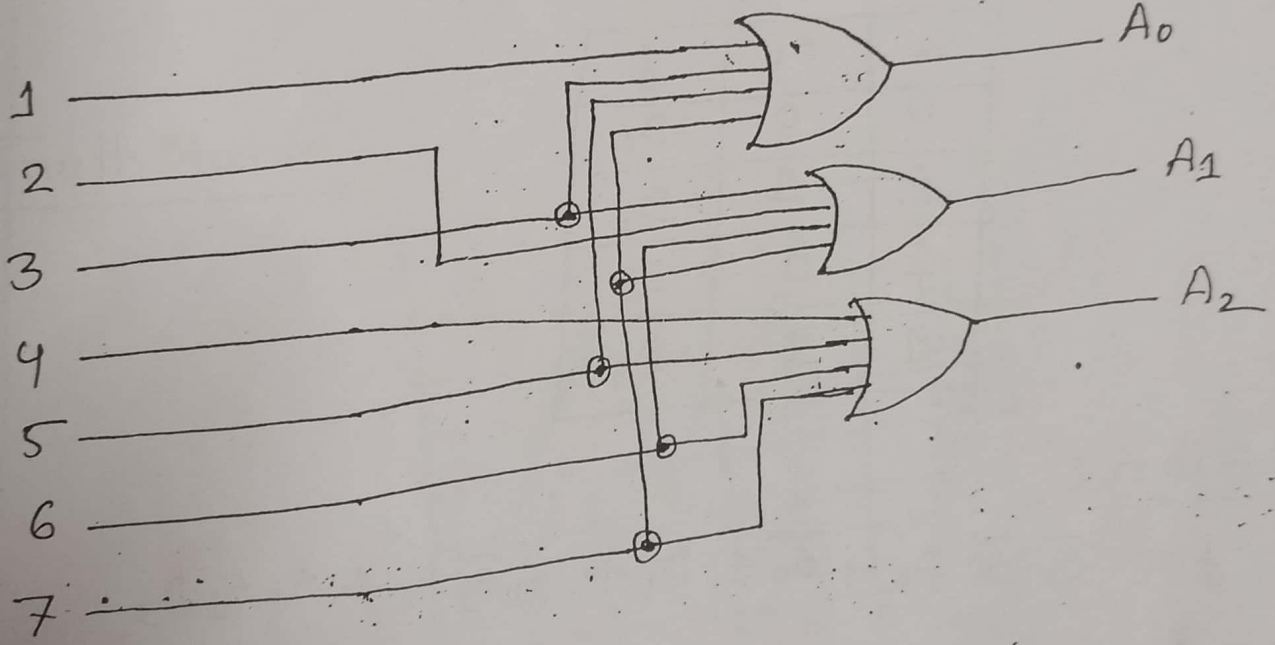
ression:

$A_0 = 1+3+5+7$.

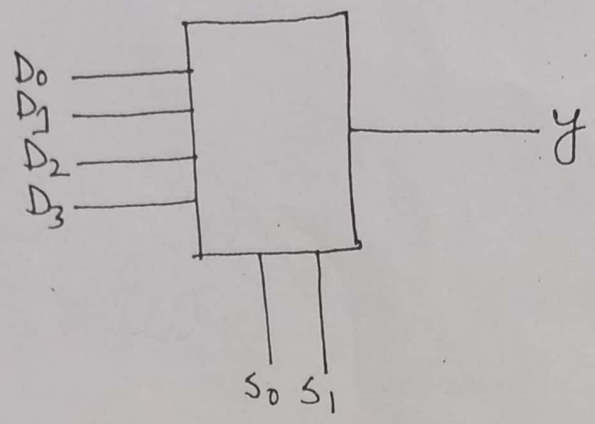
$A_1 = 2+3+6+7$

$A_2 = 4+5+6+7$.

Circuit Diagram:



* Multiplexer : 4:1 MUX



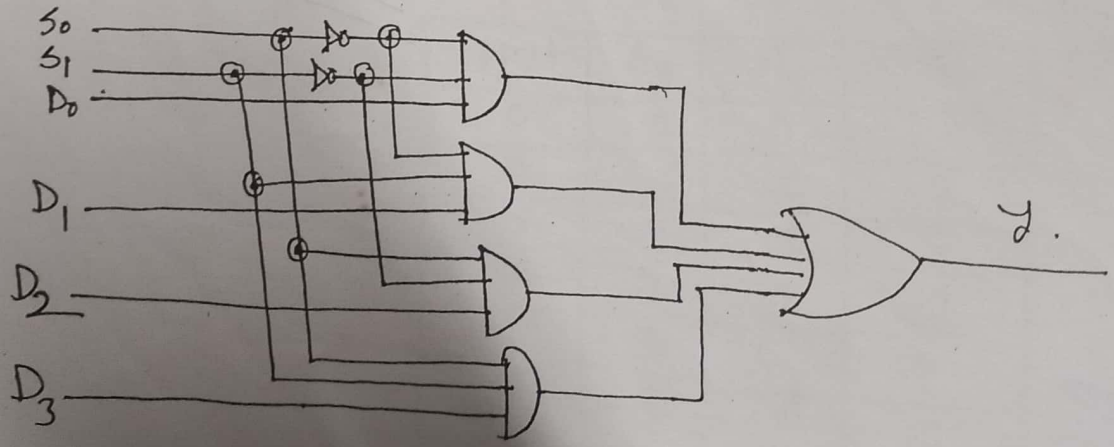
Truth table :

S_0	S_1	y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

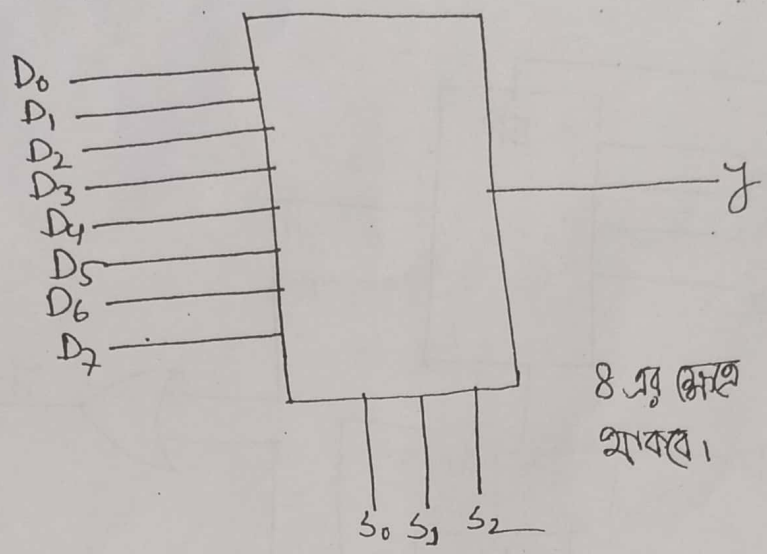
Expression :

$$y = \overline{S_0}\,\overline{S_1}\,D_0 + \overline{S_0}\,S_1\,D_1 + S_0\,\overline{S_1}\,D_2 + S_0\,S_1\,D_3$$

Circuit Diagram :



8:1 MUX



8 টা ইনপুট 3 টা Selector থাকবে।

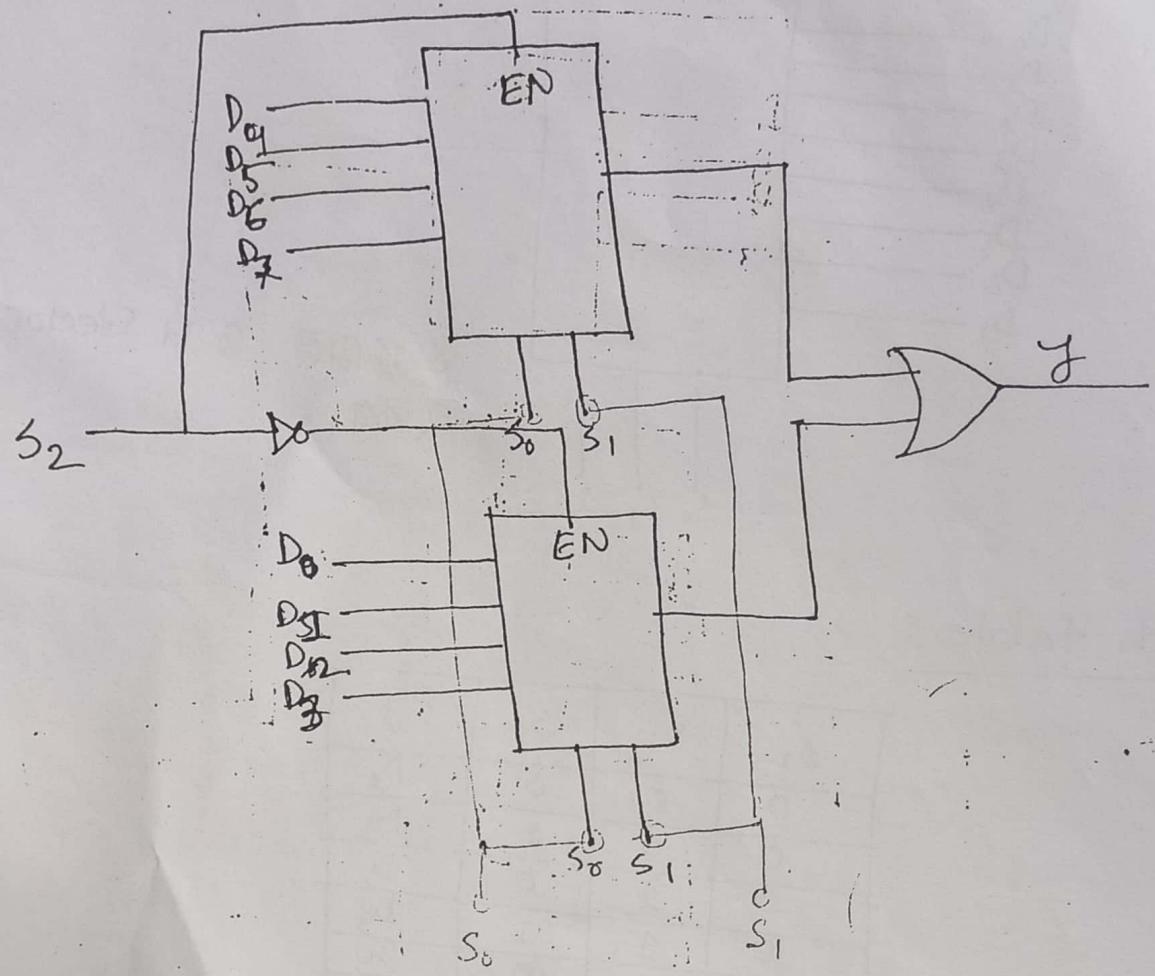
Truth table:

S_2	S_1	S_0	y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	0	0	D_6
1	1	1	D_7

$$\therefore y = \bar{S_2}\bar{S_1}\bar{S_0}D_0 + \bar{S_2}\bar{S_1}S_0D_1 + \bar{S_2}S_1\bar{S_0}D_2 + \bar{S_2}S_1S_2D_3 + S_2\bar{S_1}\bar{S_0}D_4 + S_2\bar{S_1}S_0D_5 + S_2S_1\bar{S_0}D_6 + S_2S_1S_0D_7$$

এভাবে সবগুলো Expression গুলো দেখাতে হবে।

8:1 Multiplexer from 4:1 Multiplexer

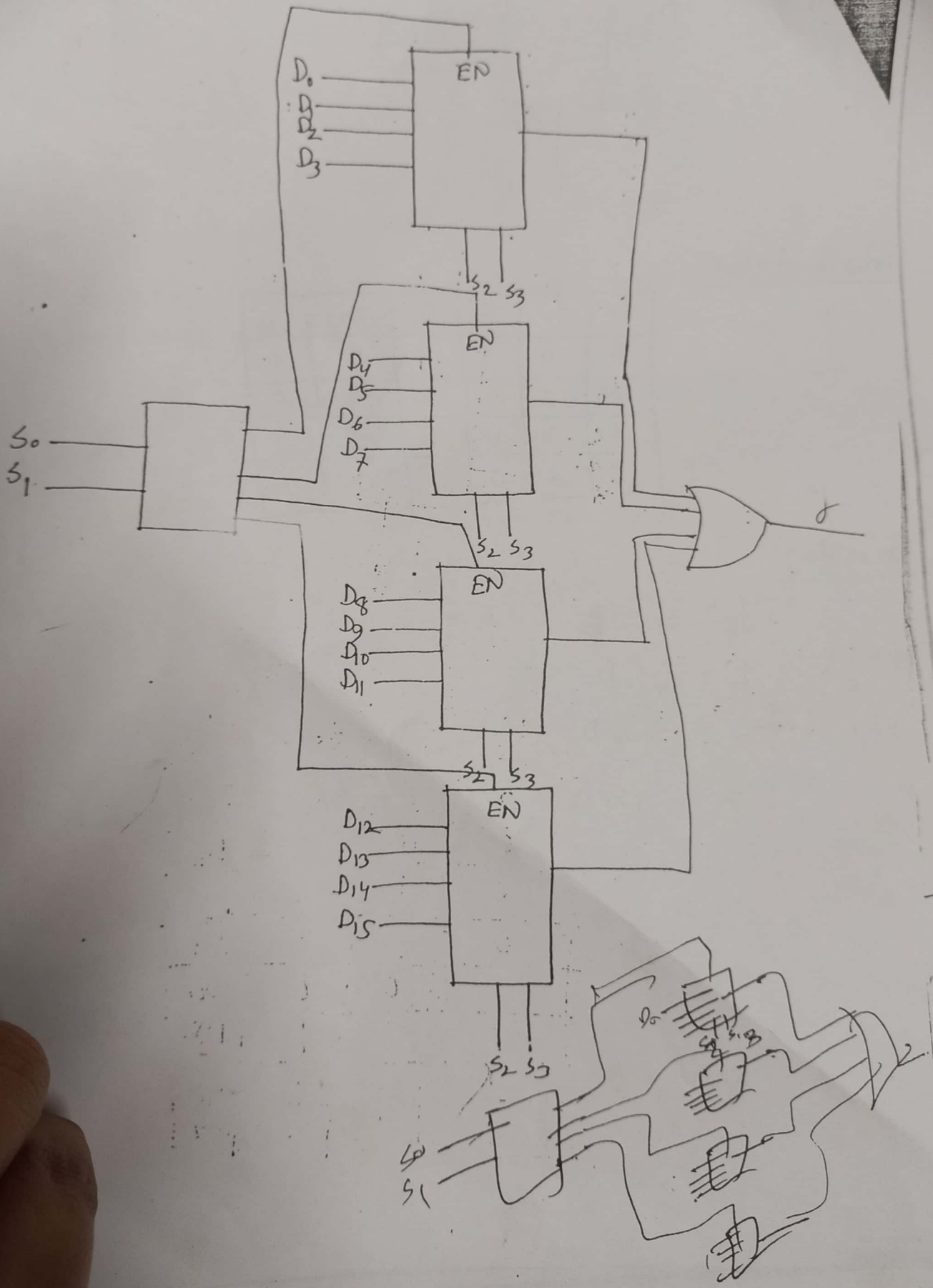


Truth table:

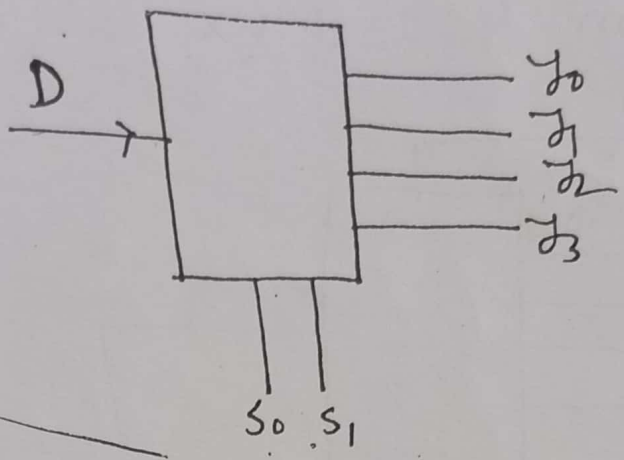
S_2	S_1	S_0	Y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

CamScanner

16:1 MUX from 4:1 MUX



Demultiplexer



Truth table:

S_1	S_0	y_0	y_1	y_2	y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Formula:

$$y_0 = \overline{S_1}\,\overline{S_0}\,D$$

$$y_1 = \overline{S_1} S_0 D$$

$$y_2 = S_1 \overline{S_0} D$$

$$y_3 = S_1 S_0 D$$

Counters

Counters: When a group of flip-flops are connected to gather to perform counting operations such a group of flip-flops are called a counter.

Basically counters are of two types based on the way they are clocked.

① Asynchronous Counter.
② Synchronous Counter.

The Difference between Asynchronous and synchronous Counters are given below:

Asynchronous	Synchronous
① In this Counter all the flip-flops are not triggered simultaneously	① In this Counter all the flip-flops are triggered simultaneously.
) They do not have a common clock pulse.	② They have a common clock pulse.
) The output of the 1st flip-flop is used as the clock for the 2nd flip-flop.	③ This counters can operate at a much higher input frequencies because of low propagation delay.
) This counters are also referred to as ripple Counters.	④ They have more complex circuitry.

either

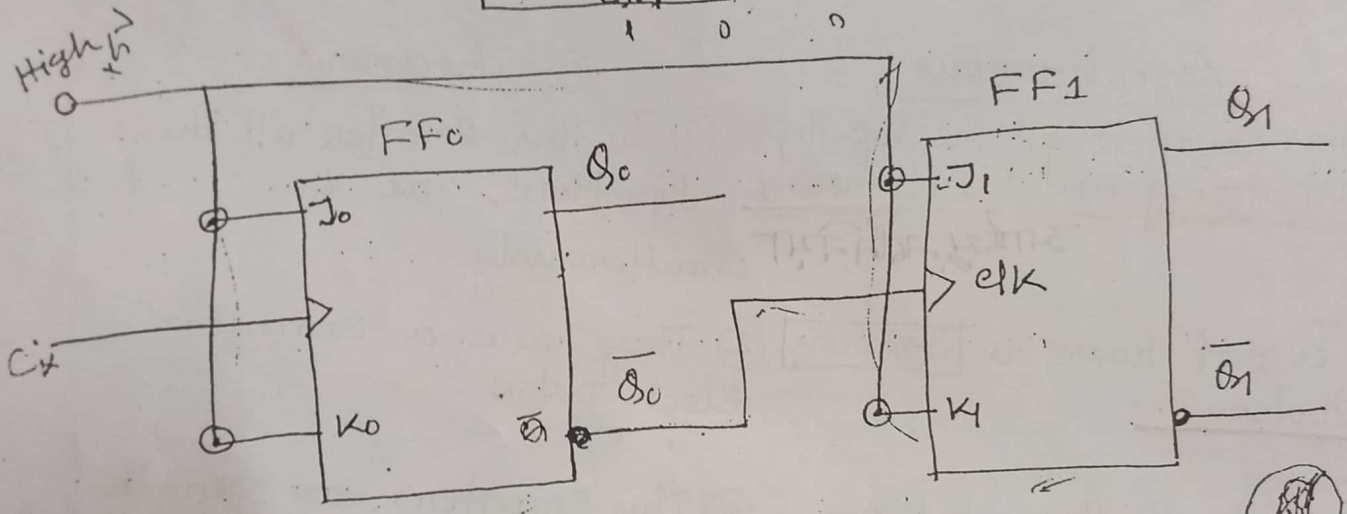
propagation delay.

Asynchronous counters

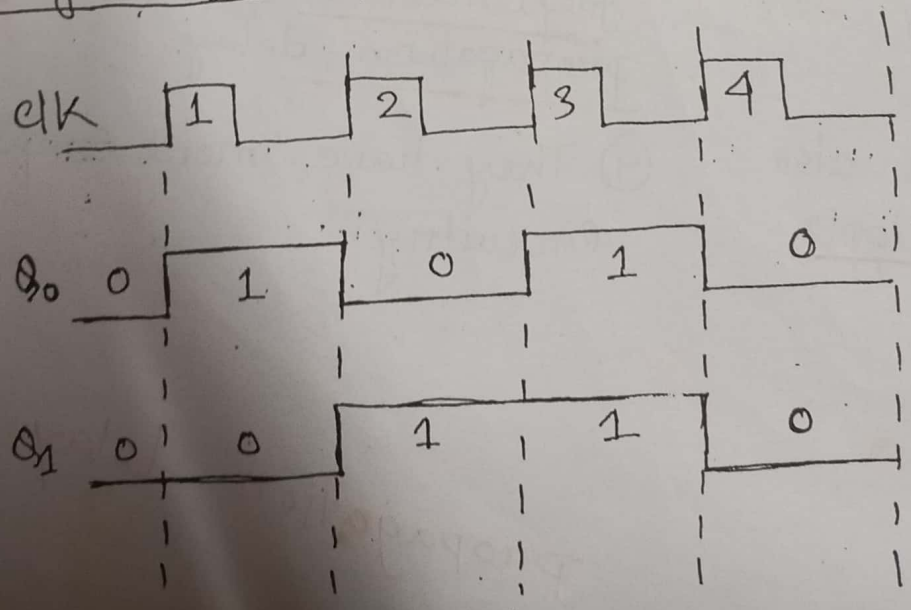
* 2-bit asynchronous counters: (MOD-4)

Truth table:

clk	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4 (re cycle)	0	0

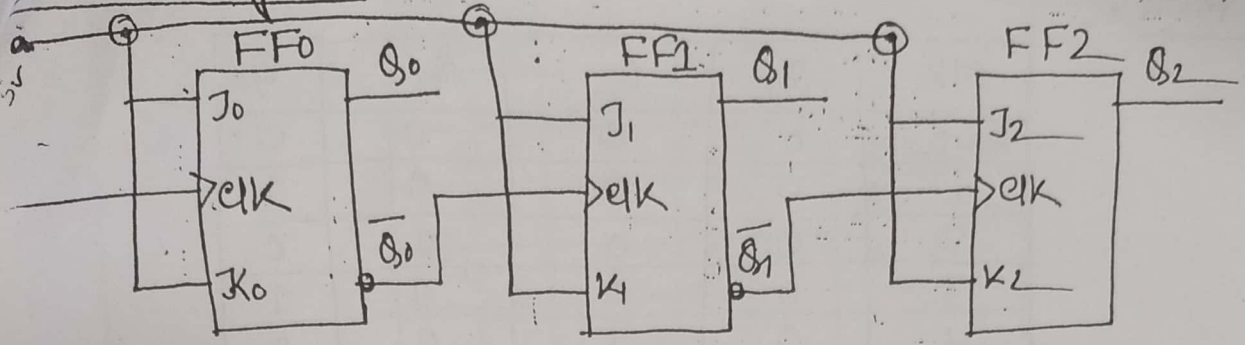


Timing diagram:



3-bit asynchronous Binary Counter: (MOD-8)

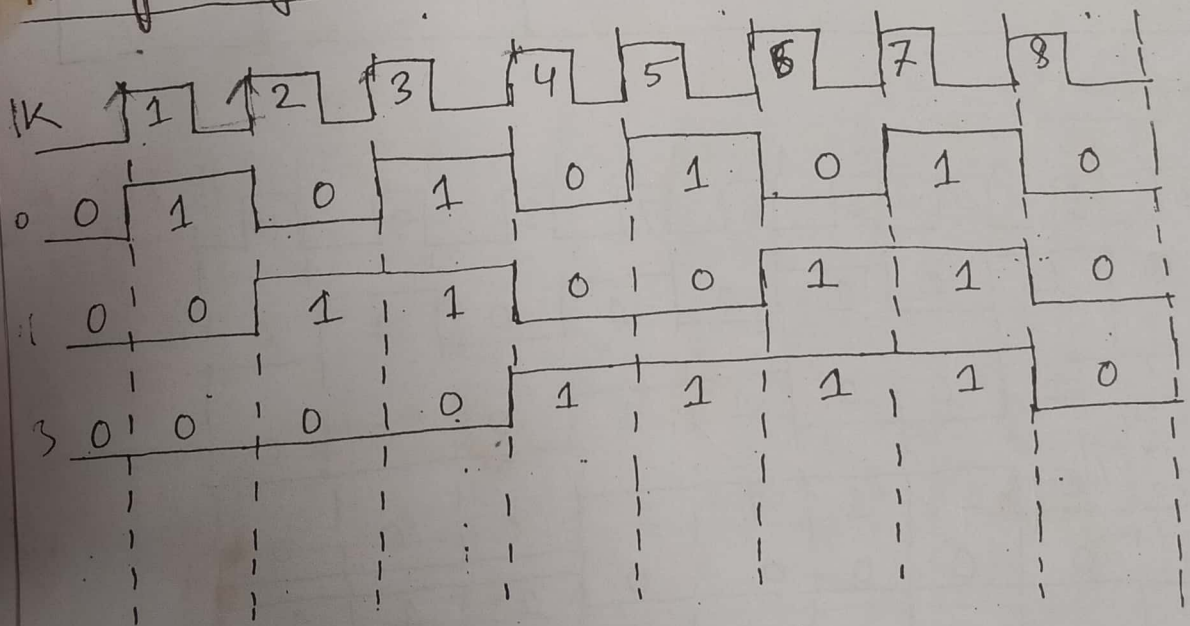
Circuit Diagram:



Truth Table:

clk	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0
	1	0	0

Timing Diagram:

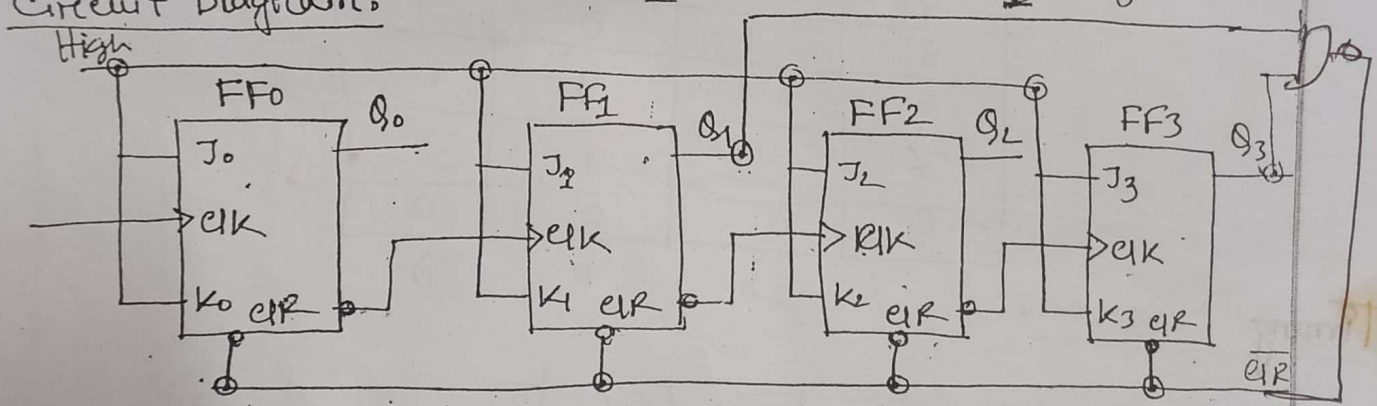


* Decade Bed Asynchronous Counter / (Mod-10):

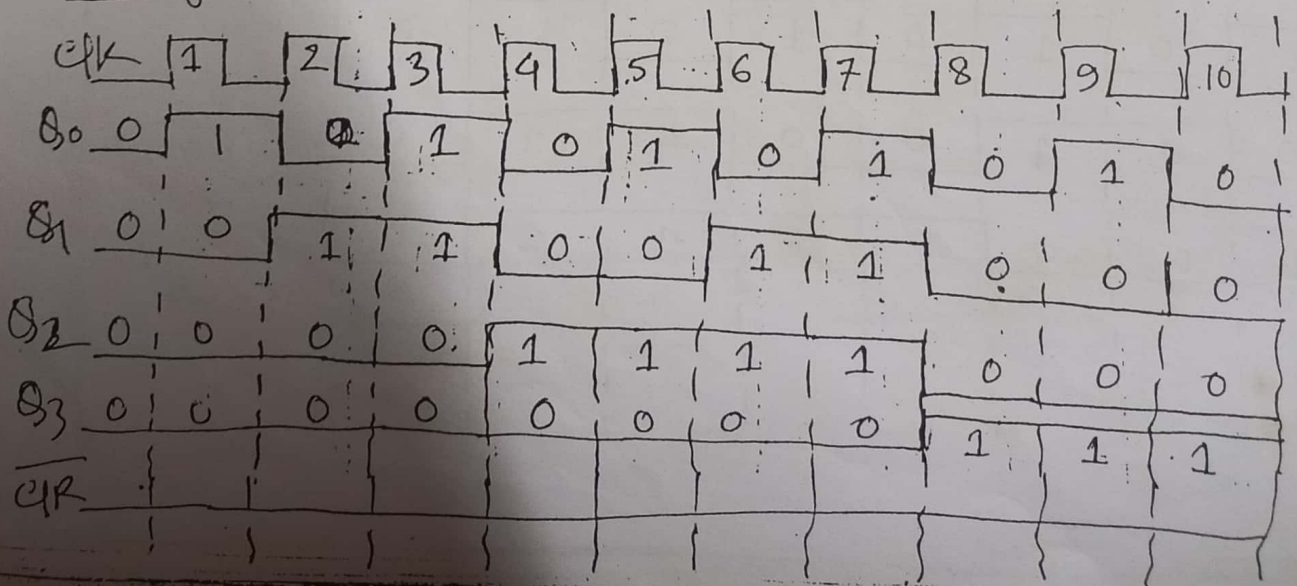
Truth table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0
	1	0	1	0

Circuit Diagram:



Timing Diagram:



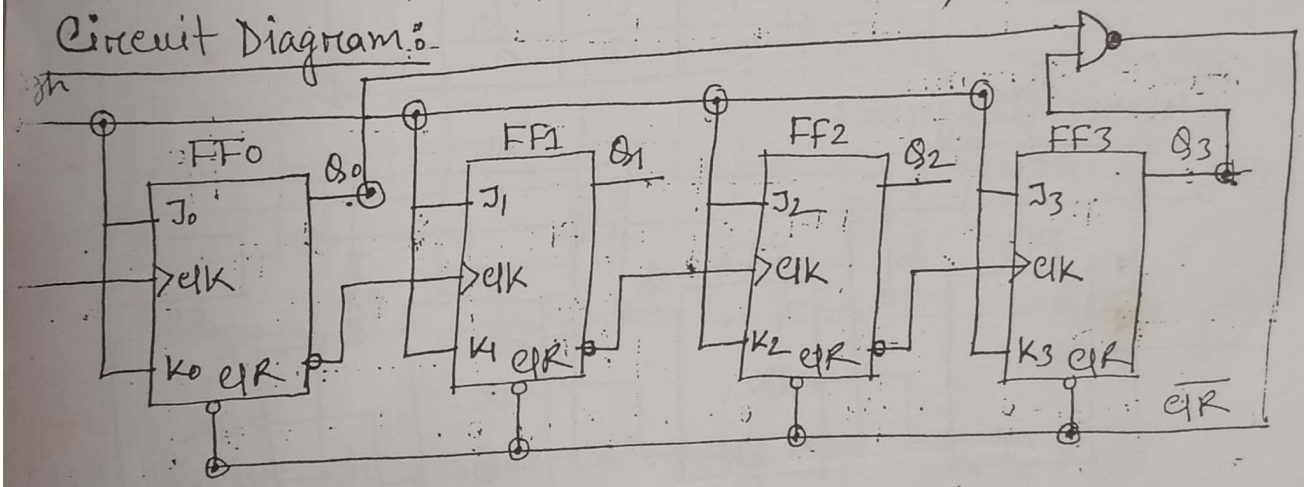
Final qus

MOD-9 Asynchronous Counter:

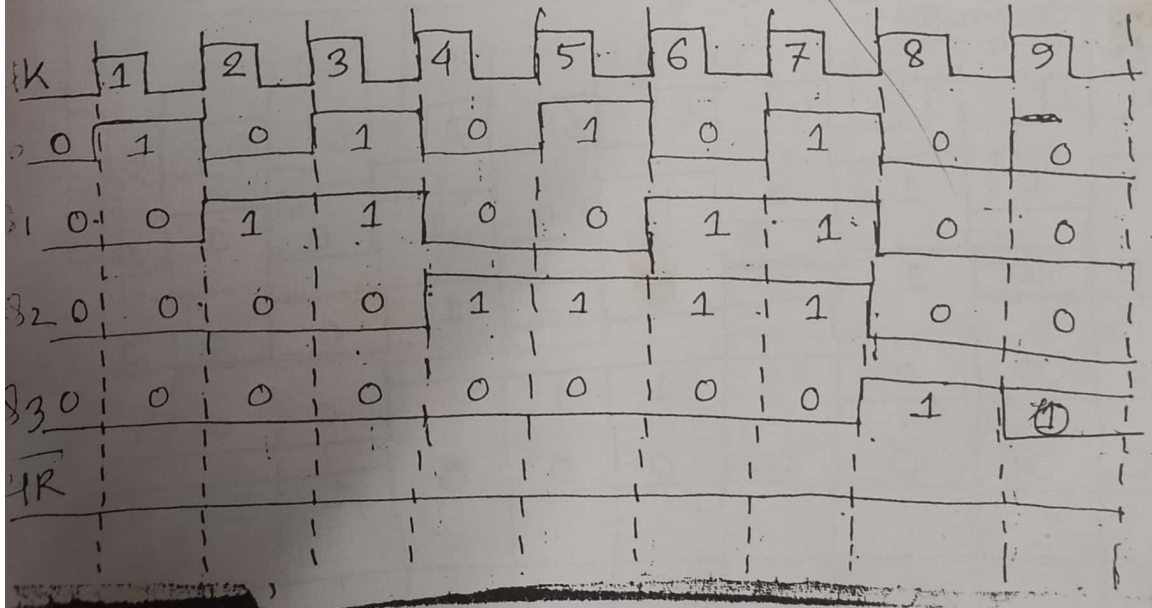
Truth table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	0	0	0	0

Circuit Diagram:



Timing Diagram:

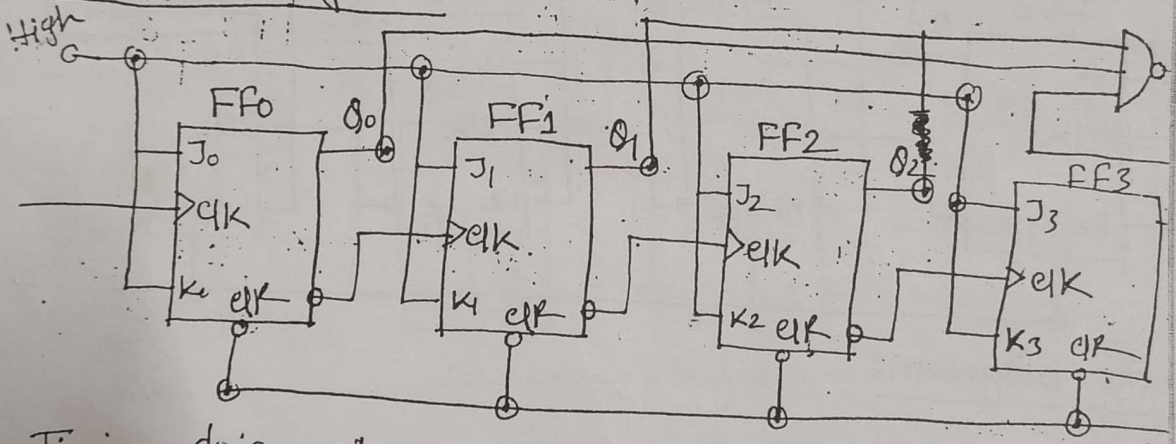


* MOD-11 Asynchronous Counter:

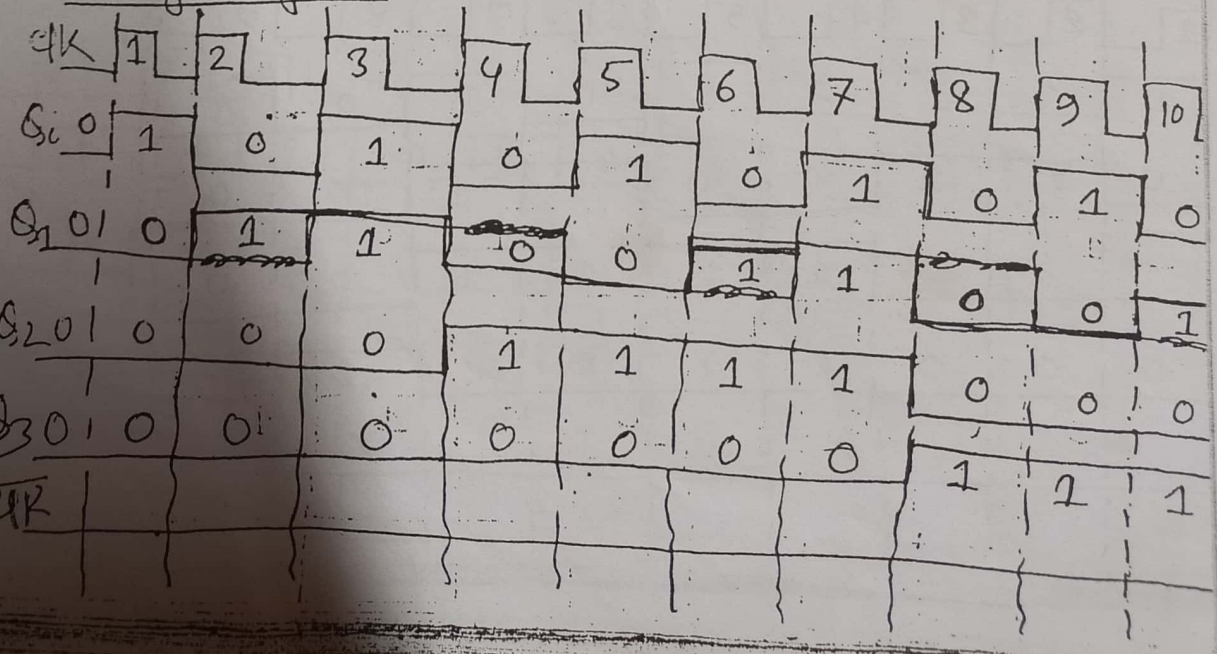
Truth table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	0	0	0	0

Circuit Diagram:

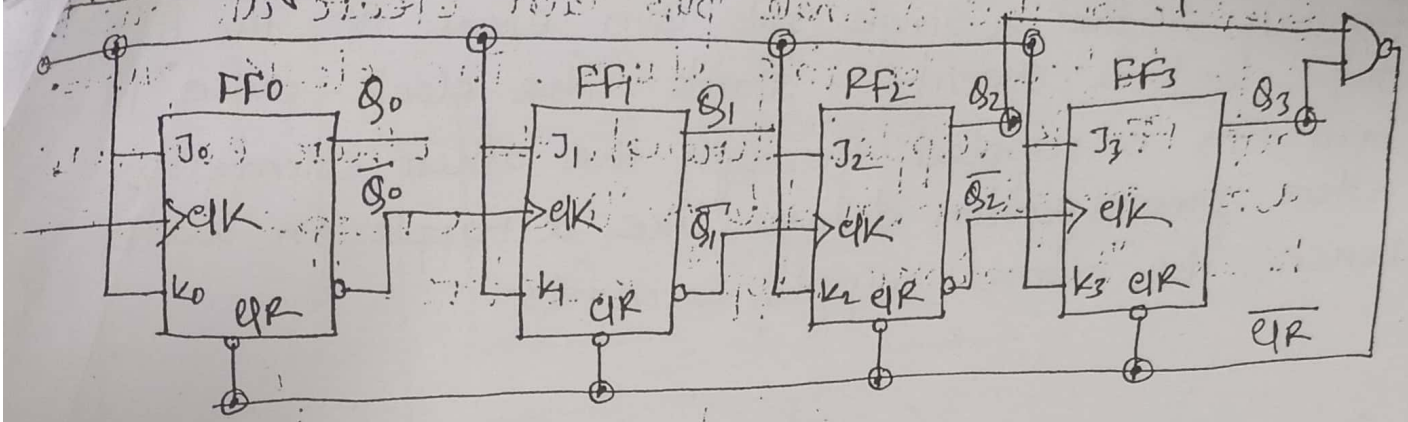


Timing diagram:

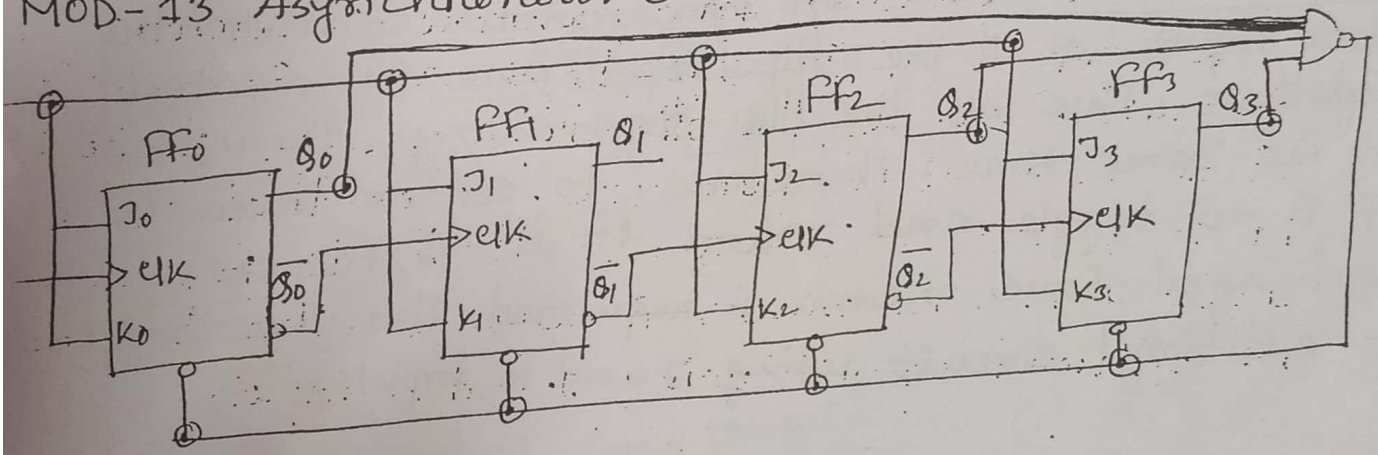


MOD-12 Asynchronous Counter:

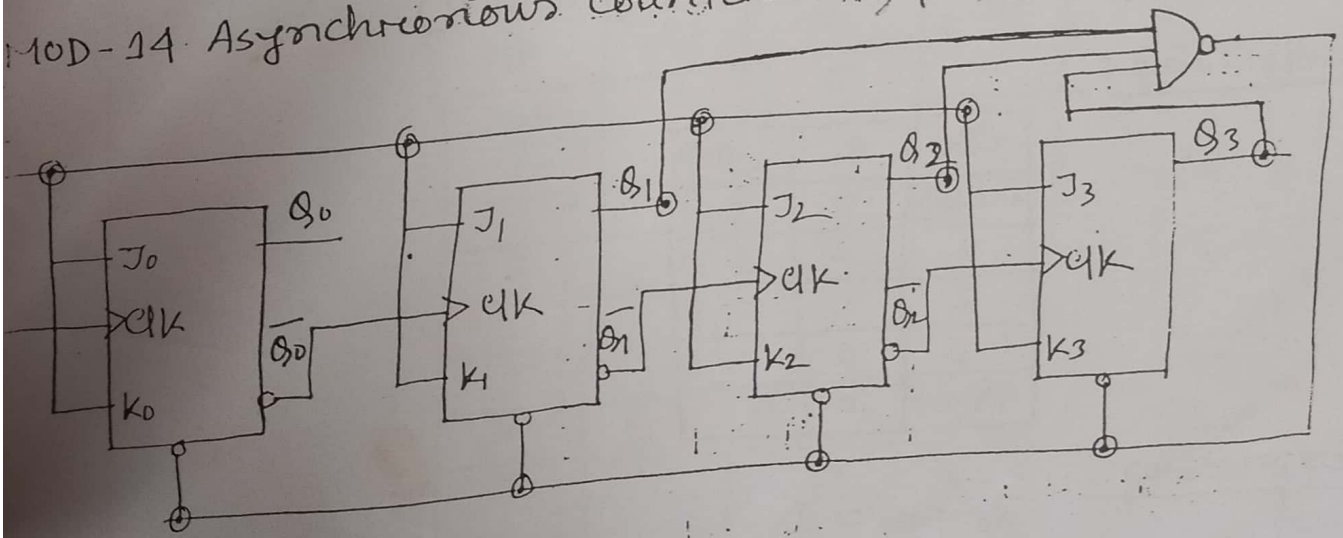
Circuit Diagram:



MOD-13 Asynchronous Counter:



MOD-14 Asynchronous counter:



Ripple Counter: Usually all the clear inputs are co... together so that a single puls can clear all the f... flops before counting starts. The clock pulse fe... into FF0 is rippled through the other counters after propagation delays like a ripple on wate... hence the name ripple counter.

Synchronous Counter

Synchronous counter: In synchronous counter the clock inputs of all the flip-flops are connected togather. Thus all the flip-flops change their sta... at the same time. Therefore, to control when a FF is to toggle and when it is to remain unchanged. Some means is required. This is accompl... by additional circuit using J and K inputs.

* 2-Bit Synchronous Counter: (MOD-4)

Truth Table:

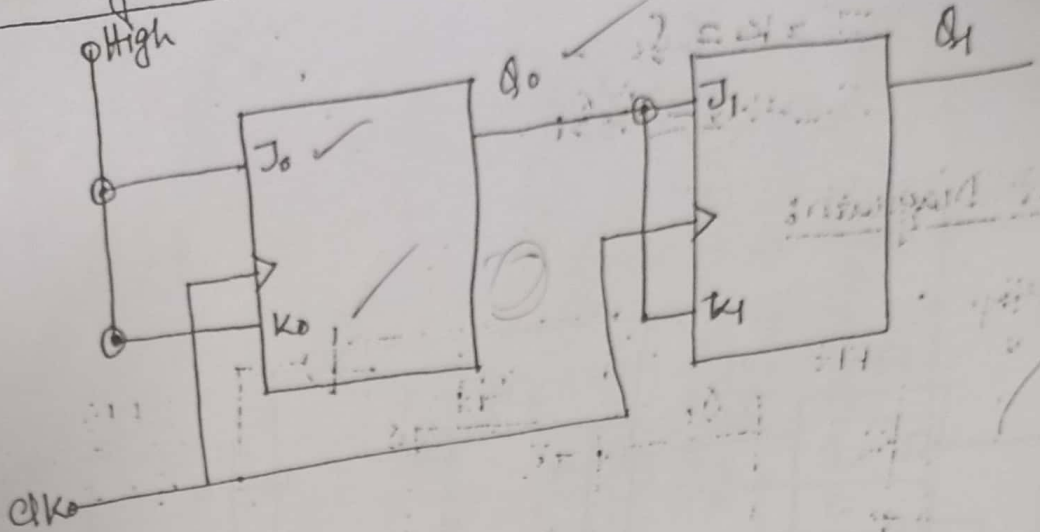
Clk	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4 cycle	0	0

Expression:

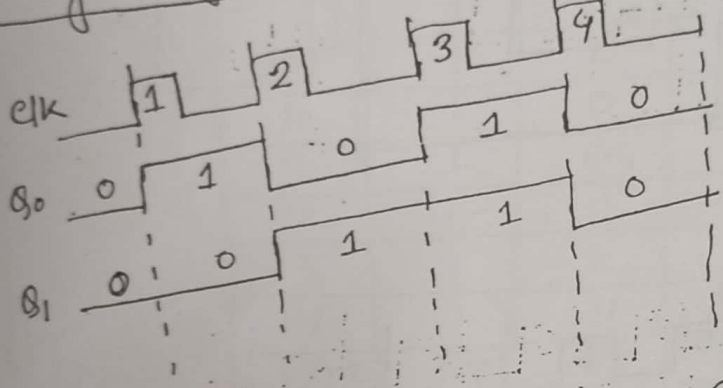
$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

Circuit Diagram:



Timing Diagram:



3-bit synchronous Counter: (MOD-8)

Truth Table:

Clk	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

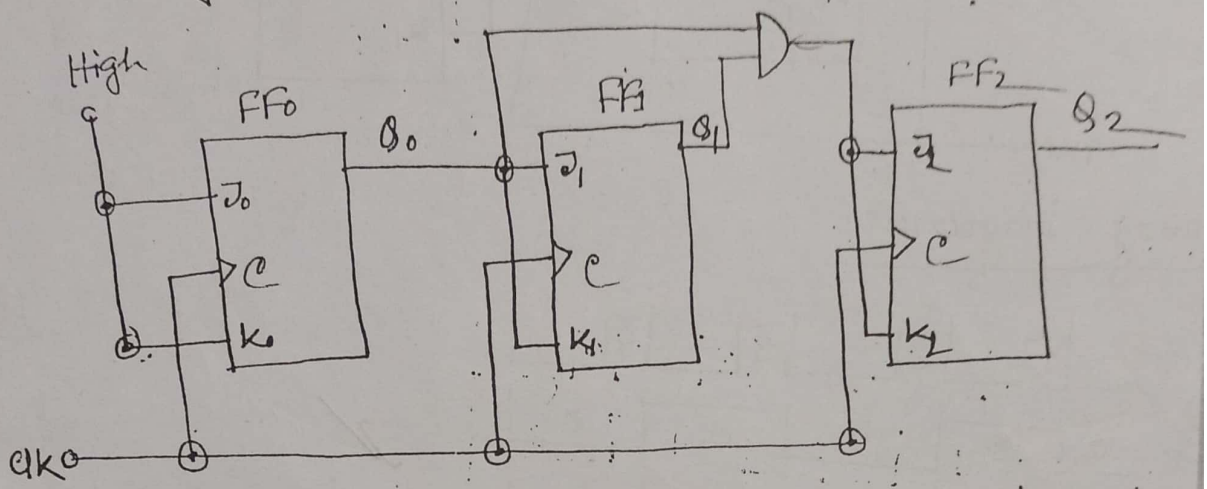
Expression:

$J_0 = K_0 = 1$

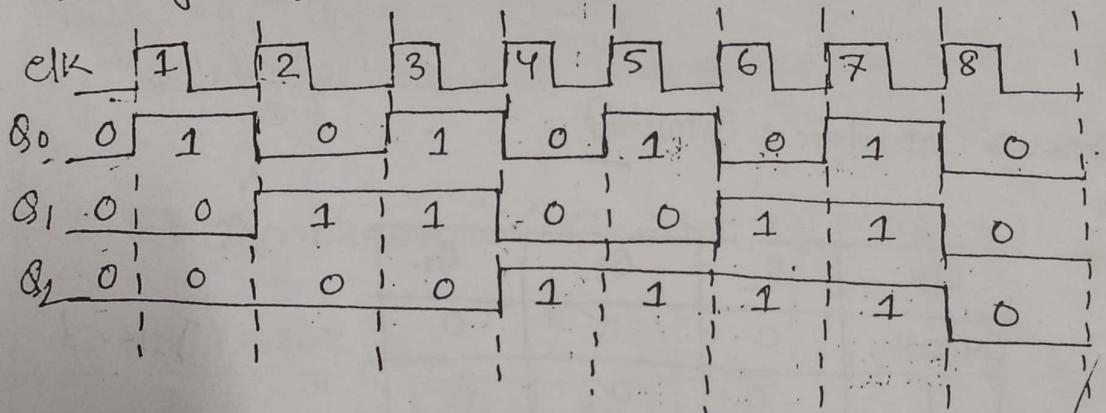
$J_1 = K_1 = Q_0$

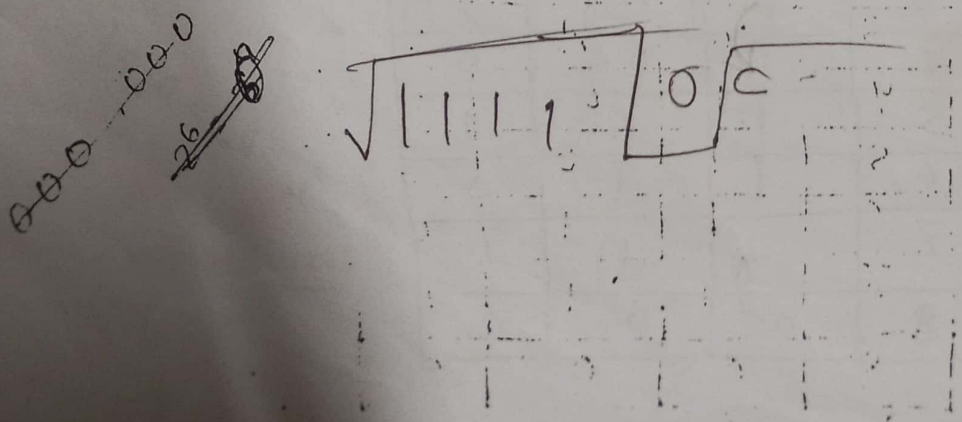
$J_2 = K_2 = Q_0 Q_1$

Circuit Diagram:



Timing Diagram:





4 bit Synchronous Counter : (MOD-16)

Truth Table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1
16	0	0	0	0

Expression:

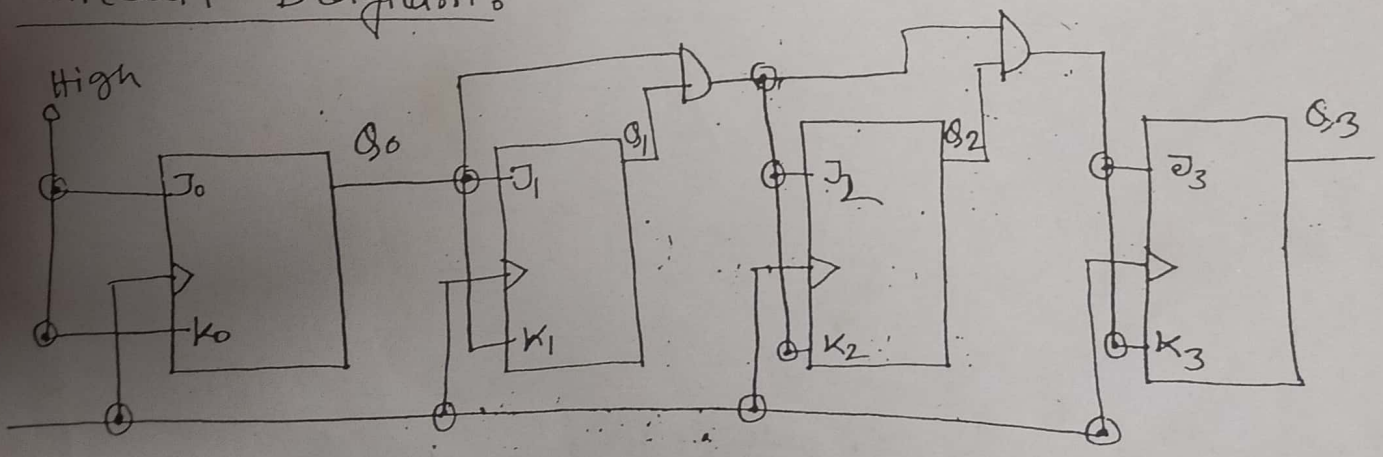
$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

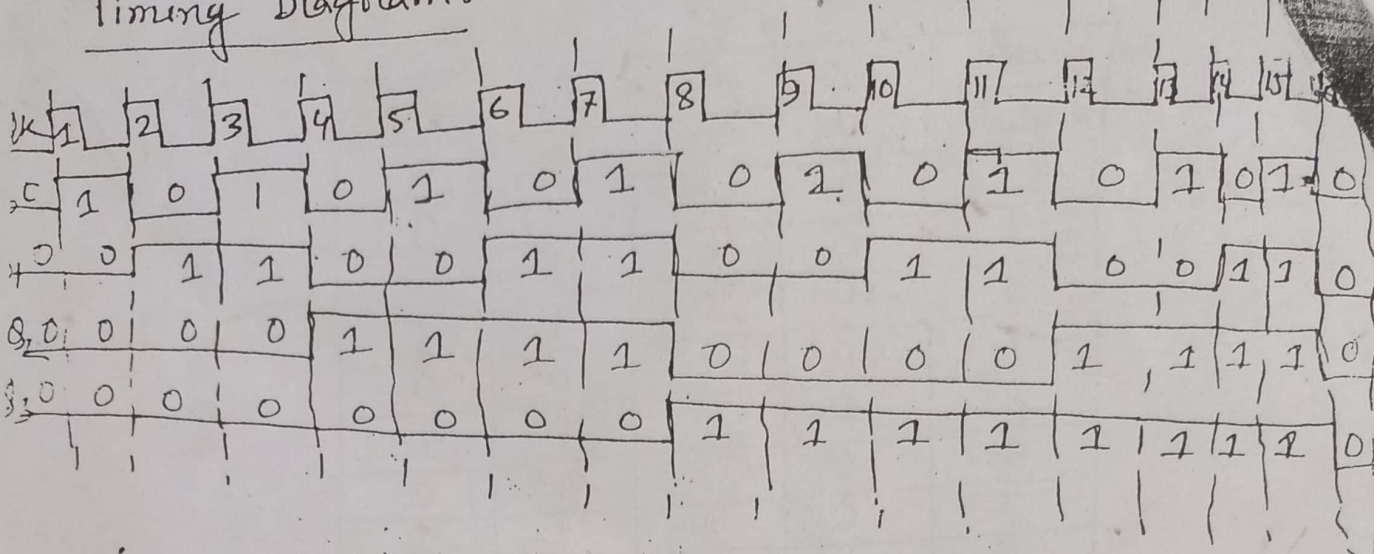
$J_2 = K_2 = Q_0 Q_1$

$J_3 = K_3 = Q_0 Q_1 Q_2$

Circuit Diagram:



Timing Diagram:



4-bit synchronous Decade counter: (MOD-10)

Truth Table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0

Q_0

Expression:

$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0 Q_3$

$J_2 = K_2 = Q_0 Q_1$

$J_3 = K_3 = Q_0 Q_1 Q_2 + Q_0 Q_3$